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The City After Abundance

Human-Centered Architectural Frameworks
for Living in Extreme Environments on Mars and Beyond

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Human-Centered Architectural Frameworks for Living in Extreme Environments on Mars and Beyond

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Abstract

Mars is often imagined as a place to begin again, a hostile yet familiar horizon where Earth's systems of extraction and exclusion can be left behind. City After Abundance resists this fiction. It asks instead: ***how do we design for life when there is nowhere left to escape to?***

Mars is the only extraterrestrial body in our solar system imagined as inhabitable rather than merely occupiable. Its day-night cycle, geological evidence of past water, and six-month proximity to Earth make it simultaneously familiar and unforgiving, an ideal site for architectural inquiry that demands permanence and collective life rather than temporary habitation. But an extraterrestrial environment like Mars requires more than technical inquiry. It demands a re-reading of architectural logic: not only in terms of performance, but in terms of what that logic has historically enabled and destroyed on Earth.

Mars intensifies consequences that Earth normalizes: territorial expansion that consumes ecosystems, extraction without regeneration, the quiet destruction of the natural systems that sustain life. In a closed and fragile habitat, these logics cannot remain unchecked. On Mars, there is no invisible labor to absorb risk, no externalized environment to receive waste. Everything is exposed. Everything matters. Architecture is no longer background. **Architecture becomes the condition of life itself, shaping how people move, relate, and endure together, directly determining individual psychology, communal relationships, and the social structures required for collective survival.**

This project does not propose Mars as a solution. Instead, it uses **Mars as a speculative lens through which a damaged Earth can be re-read and questioned**, exposing the fragilities of life here while asking how architecture can mediate care and governance when permanence and interdependence are unavoidable. Certain constants of human nature will persist across any timeline: the need for social connection, movement, nourishment, culture, and community. Designing for life on Mars means designing for these needs under radical constraint, examining how architecture can sustain psychological comfort, social cohesion, and ecological

balance when there is no margin for error.

Through speculative modeling, virtual reality-based testing and experimental prototyping, the thesis examines how, given these limitations, spaces can foster psychological wellbeing over long periods of time, and how habitats can expand while maintaining social cohesion and ecological balance. Through three scenes of everyday Martian life, at both housing and community scales, the project builds toward a prototype settlement. Not a blueprint, but a speculative ground. One possible framework among many, shaped as much by what we hope to become as by what we have not yet managed to leave behind.

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Mars does not
offer a clean start.

It offers a mirror.

What does it mean to design without the assumptions our society rests on? What do community and design owe each other when there is no outside to escape to? Mars makes these questions unavoidable, positioning architecture as an active agent in shaping life within extreme environments. Where resources are closely managed and movement is limited, architecture is no longer background; it becomes the terms of life itself.

Chapter 1

Introduction

I. Our World Today

We live in a world that promised renewal.

After the catastrophes of the twentieth century — the global spread of extractive industry, fossil-fuel modernity that made pollution and sacrifice zones feel normal, speculation-driven cities that produced luxury vacancy beside extreme housing crisis — there was a moment where reorganization felt possible.

That moment passed.



Fig. 1. Smoke billowing from a steel plant, Inner Mongolia, China, November 4, 2016. Source: Kevin Frayer, Getty Images, 2016.

What replaced it was not collapse. It was something quieter. A slow normalization. Extraction became infrastructure. Pollution entered the news cycle, and then left it, not because it stopped, but because it became unremarkable.

The violence we live with today is not ideological. It is systemic. It is the quiet decision, made again and again, about who gets to remain and who is pushed out. Who is protected and who is exposed. Climate change is no longer a fiction. Climate migration is real. And the Earth does not collapse in one moment. It frays. Inequality hardens over time. Extraction shows over time.

People are tired. And they are looking for somewhere else to go.

II. The New Destination

Mars is the answer that keeps appearing.

Worth taking seriously, not dismissing, but interrogating.

Mars is both familiar and hostile at once. Close enough to imagine life there, but harsh enough to erase the illusion that life can run on invisible forces. **It offers a rare chance to begin again, not with fantasies of perfection, but with the clarity of limits and the discipline to build from a cleaner starting point.**

But there is another reason for Mars specifically. The distance.

Enough distance to be independent. A Martian settlement cannot be governed from Earth. It cannot be resupplied on demand, or corrected from the outside. It must account for itself.

That enforced independence is not a hardship. It is a design condition. It means every system must be closed, every consequence must be local, and every decision must be made by the people who live inside it.

While on Earth, the costs of extraction are slow and diffuse, absorbed into landscapes, communities, bodies, at a scale that allows denial. The sacrifice zone is always somewhere else.



Fig. 3. Artist's impression of Mars approximately four billion years ago, depicting a vast northern hemisphere ocean. Source: ESO/M. Kornmesser, via NASA Astrobiology, 2025.

On Mars, that denial is impossible. No frontier to expand into. No atmosphere to absorb waste. Every act of taking must be accounted for. The extraction economies that have structured life on Earth become legible here precisely because they cannot hide.

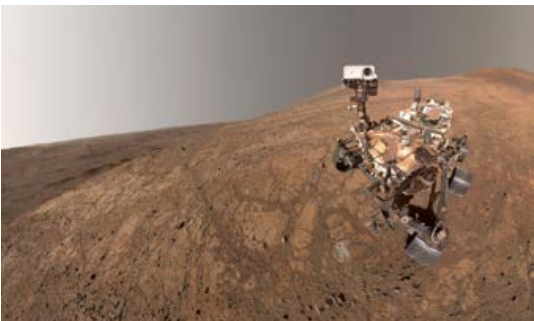


Fig. 2. NASA's Perseverance rover selfie on its 1,500th Martian day, May 10, 2025. Source: NASA/JPL-Caltech/MSSS, 2025.

Chapter 2

Thesis Typology

City After Abundance does not begin with buildings. It begins with infrastructure. On Mars, survival precedes form. Before a space can exist, there must be water, energy, and the ability to move between them.

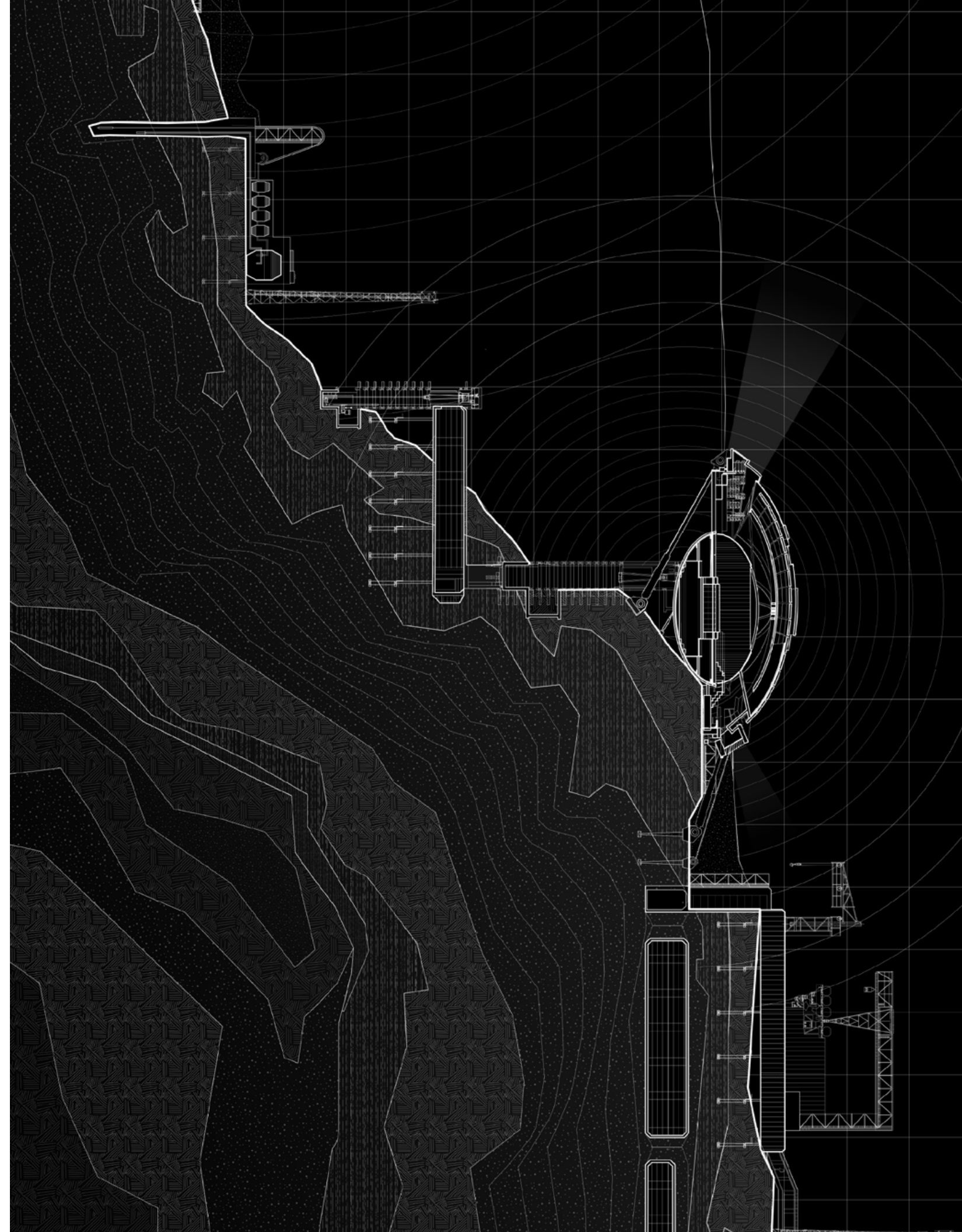
Situated within Utopia Planitia, the largest recognized impact basin on Mars, where subsurface ice and minerals are the primary resource, the city organizes around interdependent systems: ice mining hubs, processing and distribution networks, and transportation nodes. Habitation grows from them, extends between them, and deepens within them.

The architecture is carved into the ground rather than placed on it. **Excavated volumes respond to radiation and temperature extremes, but they also produce a new spatial language, one of scale, depth, gradients of light, and continuous interior landscapes.** Architecture becomes something you not only move through, but feel and live with.

Scale differentiates experience. Some spaces make the body feel small, like the communal dining hall and the meditation room, emphasizing dependency on larger systems. Others hold the body: the infrastructure hub, the hydroponic garden, the individual living quarter. Together they constitute a full range of inhabitation.

This is a city for 2500. But it is not a prediction. not a blueprint. It is a prefigurative act, using an imagination to make visible what the present refuses to see.

Fig. 4. Section through City After Abundance. Source: Author.



Chapter 3

Historical Context

I. Speculative Architecture as Critique

Speculative architecture has long been used as a tool to critique the systems of its time. In moments of insecurity, especially during crises or periods of transition such as the aftermath of wars and the industrial revolution, architects turned imagination into a method of questioning, not to propose what should be built, but to make visible what already existed.

New Babylon imagined a post-labor world of collective creativity and spatial freedom. Designed by Constant Nieuwenhuys in response to the rigidity of postwar urban planning, it proposed a world where people were no longer tied to work or place, drifting instead through interconnected environments shaped by play, chance, and collective desire.

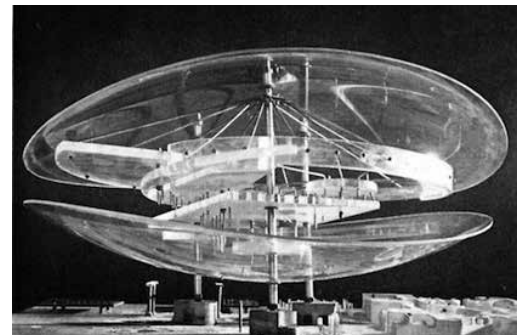


Fig. 5. Constant Nieuwenhuys, New Babylon—Concert Hall for Electronic Music, 1958–1961. Source: Verso Books, 2016.

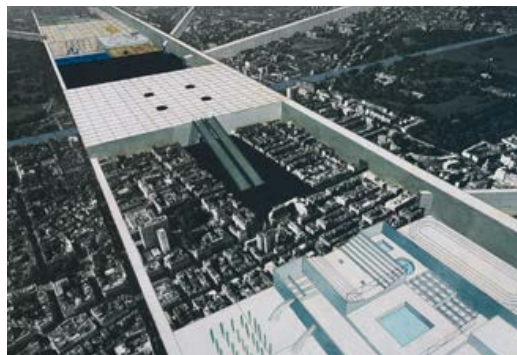


Fig. 6. Rem Koolhaas, Elia Zenghelis, Madelon Vriesendorp, and Zoe Zenghelis, Exodus, or the Voluntary Prisoners of Architecture: The Strip, 1972. Source: MoMA, 1972.

Exodus, or the Voluntary Prisoners of Architecture, revealed architecture as a system of control, seductive, totalizing, and never neutral. Rem Koolhaas and Elia Zenghelis imagined a walled strip cutting through London, a space so intensely desirable that people would choose to enter and never leave. The project exposed how architecture does not simply house power, it produces it, shaping behavior and desire from the inside out.

Archigram's Walking City and Plug-In City challenged fixed geography by proposing mobile, replaceable infrastructures. By imagining cities as machines that could



Fig. 7. Ron Herron, Walking City Visiting New York, 1964. Source: Ron Herron, 1964.

assemble, disassemble, and relocate, they dismantled the assumption that architecture was stationary.

Archizoom's No-Stop City stripped urbanism to its bare logic. A continuous, undifferentiated interior, endlessly repeated and devoid of hierarchy. It did not celebrate this condition but exposed it, making visible what a system built on growth without purpose actually produces: consumption without meaning, space without life.



Fig. 8. Archizoom Associati, No-Stop City, 1969. Source: Archizoom Associati, 1969.

Speculative architecture has historically functioned as critique, exposing the hidden logics of existing systems by following them to their limit. But critique alone is not enough when the systems being critiqued are already visibly failing. The distance that once allowed denial is collapsing.

This is where this thesis departs from its precedents. These precedents held a mirror up to the present. This project does something different. It uses that extreme conditions of another world to test a way of living that the present world has not yet allowed. Not as prediction. Not as escape. As rehearsal to see what it's like to build a city from ground-up.

II. The Normalization of Crisis

The project takes root in a century shaped by industrial sprawl, fossil fuel dependency, and a relentless pursuit of growth — one that came at the expense of social equity and ecological health. The post-war promise of collective progress proved self-defeating: rather than lifting public life, it accelerated the very forces of resource extraction and financial speculation that hollowed it out, concentrating wealth in the process.



Fig. 9. Human activity and drought degradation in the Amazon rainforest. Source: Douglas Magno, AFP/Getty Images, 2024.

By the late twentieth century, urbanization deepened the divide between those who profited and those who were displaced. **Cities learned to mask their true cost, relying on invisible labor and distant resource frontiers to sustain the economy.** The system depended on the persistence of poverty and exploitation.

The warnings came early. Scientists, communities, and activists named what was happening. There was a moment when those warnings felt urgent. Then they stopped appearing on the news, not because it stopped, but because it became the so normalized.

Climate change was discussed as a future problem, then treated as an inconvenience, then absorbed into daily life as simply the way things are. Climate migration followed the same pattern. People were pushed from places they once called home, moving quietly and in large numbers toward places that did not welcome them. Without our knowledge, the sacrifice zone was moving closer.

This normalization of the problems in our world today is not passive. It is produced over time, through the active suppression of urgency, the repeated deferral of consequence onto those least responsible, and the design of cities and institutions that keep the cost of extraction invisible to those who benefit from it.

III. The Question Nobody Asked

Space exploration first captured the public imagination often in response to periods of insecurity, whether political, environmental, or ethical. After wars, depressions, and economic collapse, the ability to imagine exploring new territory raised urgent questions: **What would happen if Earth failed? What if we exhausted its resources?** The prospect of living in outer space became both a form of escape and a catalyst for new ways of thinking.

The idea of building beyond Earth has been part of the conversation for decades, **but it began not as a question of how humanity might live differently, but as a demonstration of power.** Each mission was a symbol, a flag planted, a border drawn in orbit. The race to space was never really about human experience. It was about who could project dominance furthest from home.



Fig. 10. Artist's impression of the interior of a Stanford Torus space habitat, depicting a large-scale rotating cylindrical structure designed for permanent human habitation in space. Source: Don Davis/NASA, 1975.



Fig. 11. Cold War-era rocket development, United States and Soviet Union. Source: NASA.

And yet, buried inside that competition, a genuine question went unasked. If we were to build a city beyond Earth, what kind of city would it be? Would it carry forward the same systems, the same hierarchies, the same logics of extraction and exclusion, the same structures that produced sacrifice zones on Earth? Or would the distance, the constraint, the absolute necessity of cooperation, force something different?

IV. This Thesis

This thesis refuses that assumption. It builds on the lineage of speculative architecture and takes that critique one step further.

Mars is a site where the systems that Earth’s abundance allowed us to ignore can no longer remain hidden. Where there is no elsewhere to externalize the cost. **Where the question of how we build is inseparable from the question of how we live together.**

Mars is the instrument, not the subject. Its value is its hostility: a place where consequence cannot be deferred, where every system must close its own loop, where the distance we built into our current infrastructures to make extraction invisible is structurally impossible. **That constraint is not a limitation on the speculation. It is what makes the speculation honest.**

Chapter 4
Theoretical Context

This thesis goes further than exposure.

The method is prefiguration — representing, imagining, or foreshadowing something in advance, as a prototype for future ways of living. **It is the practice of something before it exists, so that when the conditions arrive, the imagination is already there to meet them.**

To ask how we should build our world from the beginning is to transform speculative critique into an act of repair. An architecture that grows naturally, not imposed by existing systems, but shaped by those who inhabit it.

The question, then, is not only: what is wrong with the world we have?

It becomes: what would it feel like to live differently?

The theoretical ground beneath this project is the critique of extractivism — the logic by which land, labor, and atmosphere are treated as resources to be consumed rather than systems to be sustained. On Earth, extractivism operates through distance. The sacrifice zone is always somewhere else. The cost is always absorbed by someone not in the room where the decision is made. Architecture participates in this logic when it conceals its systems, separates people from the infrastructure that serves them, and treats consequence as someone else’s problem.



Fig. 12. Aerial view of a mining site with heavy machinery. Source: Environmental Defense Fund.

Mars removes that distance. With no frontier to expand into and no atmosphere to absorb waste, every decision becomes local and every cost becomes visible. The extreme environment does not produce a new problem. It reveals an old one.

Chapter 5

Project Justification

Architecture has always been political, yet the discipline has largely retreated from acknowledging it. The environments we inhabit shape our consciousness, behavior, and the conditions through which collective life becomes possible. But in a moment of accelerating ecological collapse, deepening housing inequality, and the normalization of sacrifice zones, architectural discourse continues to privilege formal innovation and speculative futures that appear appealing on the surface while leaving systems of extraction untouched.

Numerous competitions have imagined life on Mars and the Moon through inflatable habitats, robotic construction, and autonomous survival systems. **Most frame extraterrestrial architecture primarily as a technical and engineering challenge to be solved through efficiency, optimization, and survival.** While these concerns are necessary, such proposals rarely question the kinds of social relations, political systems, or ethical frameworks embedded within the spaces they produce. Architecture becomes infrastructure for survival rather than a medium through which collective life is shaped.

The gap this project addresses is specific. While it still assumes that architecture as a necessary infrastrure for survival, it also presents it as a way to redesign the assumptions of the llife itself. There is little architectural work that uses extreme environments as ethical mirrors rather than engineering problems. **Most proposals for extraterrestrial habitation reproduce the same extractive and hierarchical spatial logics that have failed on Earth, treating Mars as another frontier to conquer rather than a condition demanding an entirely different set of values.** Questions of governance, dependency, labor, maintenance, and care are often secondary to technological performance.

By situating the critique on Mars, the project makes visible what Earth has normalized. In a closed, resource-limited environment, the hidden costs of how we build and govern become unavoidable. There is no invisible labor to absorb waste

Given what we know from Earth,
how do we design a future that
does not repeat it?

and no elsewhere to externalize harm. Every decision about water, energy, food, and space carries immediate collective consequences. Mars therefore becomes less a destination than a lens through which to confront the assumptions embedded within contemporary life on Earth.

The contribution is both methodological and disciplinary. The project continues the architectural tradition of speculative design as critical practice, using an extreme spatial proposition not as a blueprint for colonization but as a diagnostic tool for the present. It argues that architecture’s role in conversations about planetary futures should not be to optimize corporate visions of expansion, but to contest them: to ask what kind of life, governance, and care we choose to embed in the spaces we design.

Chapter 6

Precedents

This section examines precedents across three scales—building, city, and extraterrestrial systems—to understand how architecture organizes movement, social interaction, and collective life. Rather than focusing on form, it traces how circulation and spatial structure operate as behavioral systems, from therapeutic environments and social housing, to early urban models, to space habitation strategies under extreme environmental constraints. Together, these precedents establish architecture as a framework that regulates experience through movement, density, and environmental embedding.

I. Architecture of Care

The Paimio Sanatorium by Alvar Aalto

The Paimio Sanatorium serves as a significant precedent for this thesis because it conceives of architecture as an active participant in human well-being rather than merely a container for habitation. Designed for tuberculosis patients within a pine forest in Finland, the project demonstrates how spatial, environmental, and sensory conditions can be orchestrated to support both physical and psychological health.



Fig. 13. Generous terraces on the Paimio Sanatorium providing patients access to fresh air. Source: Gustaf Welin/Alvar Aalto Museum,

Aalto’s integration of the natural landscape into the healing environment is particularly relevant to a Martian context where direct access to ecological systems is absent. The sanatorium’s terraces, sun balconies, forest views, and walking paths provided patients with daily contact with the surrounding environment. In a Martian settlement, architecture must similarly create opportunities for meaningful

engagement with natural processes through cultivated landscapes, communal gardens, environmental simulations, and collective outdoor spaces within protected habitats.

Aalto also focused on the lived experience of the individual patient through the perspective of the reclining body. This attentiveness to embodied perception carries particular relevance on Mars, where reduced gravity would alter how inhabitants move through space and how architectural conditions including ceiling heights, light angles, and spatial compression register physically and psychologically. The sanatorium's careful calibration of light, acoustics, color, and comfort suggests an approach to Martian habitation that prioritizes well-being alongside technical performance.

Communal spaces were equally designed to sustain social life within an isolated building, with Aalto treating collective space not as an addition but as integral to the healing process itself. The building recognizes that recovery is not only a bodily condition but a social one.

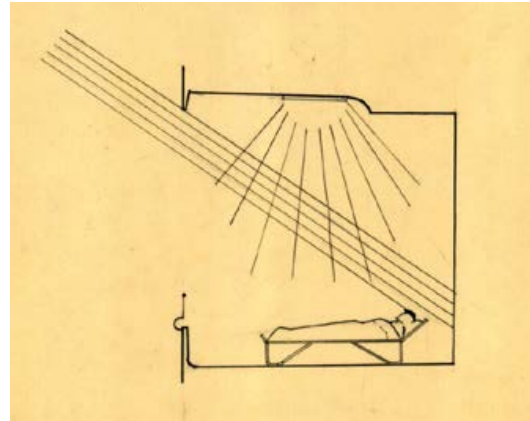


Fig. 14. Sketch showing the use of natural light, Paimio Sanatorium. Source: Alvar Aalto Foundation.

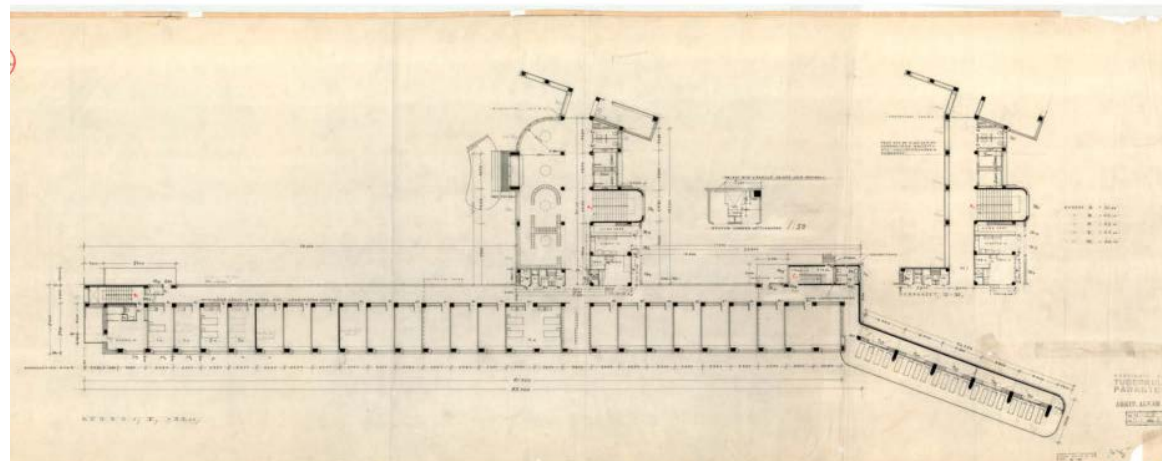


Fig. 15. Plan of patient wing, Paimio Sanatorium. Source: Alvar Aalto Foundation.

What matters most for this thesis is the way Aalto treated the act of moving through a building as an experience worth designing for. Corridors are carefully proportioned and naturally lit, their character modulated through changes in light, color, and

compression, transforming passage into a regulatory system for both bodily and psychological states.

This understanding of circulation as behavioral infrastructure has informed contemporary housing and rehabilitation design, where corridors, staircases, and thresholds are no longer merely routes but extensions of domestic life. These transitional spaces mediate between privacy and collective life, producing moments of encounter, pause, and orientation, a spatial continuum that allows inhabitants to modulate their own exposure between withdrawal and connection.

Circulation is not infrastructure in the background of life, but a behavioral framework at its center, structuring rhythm, encounter, and the architecture of belonging.

The Social Housing Projects of Álvaro Siza

Where Aalto locates behavioral support within the carefully designed interior spaces, Álvaro Siza extends this logic into the collective fabric of housing itself. His social housing projects in Porto demonstrate how circulation can become the primary site of community life rather than its byproduct.

At the Bouça Housing complex, long exterior gallery corridors act like elevated streets. Residents pause, meet, and appropriate them as extensions of home. The line between private units and shared space is intentionally soft.

At Malagueira, an infrastructural spine organizes the settlement. It carries both movement and services. Circulation and infrastructure merge into one system that shapes daily rhythm as much as spatial order. By negotiating the gradient between public street and private domesticity, the architecture utilizes a series of courtyards and direct thresholds to dissolve traditional boundaries. This blurs the boundary between street life and domestic space and encourages informal social interaction.

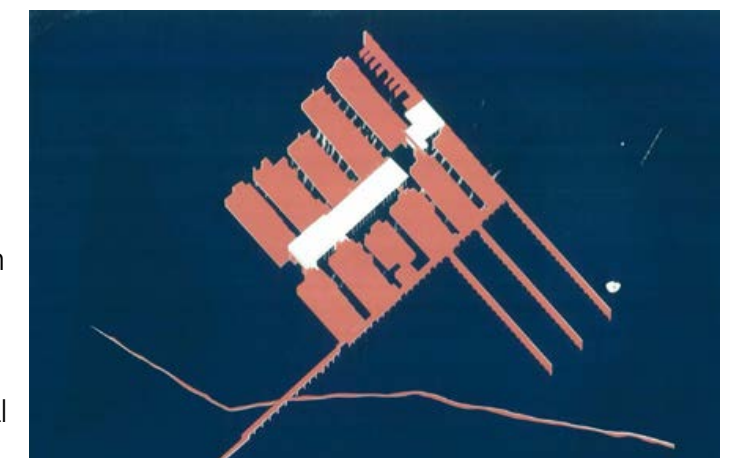


Fig. 16. Catherine Shiner, Malagueira and Évora, Portugal, 2016. Source: Drawing Matter, 2016.

For a Martian settlement, where the boundary between interior and exterior is defined by survival, Siza's work suggests that collective passage is not secondary, but foundational. Corridors and shared thresholds become the armature that structures habitation and makes collective life possible.ecture of belonging.

The Framed View of Mies van der Rohe

Continuing the idea of framing the view to nature outside from Aalto's sanatorium, Mies van der Rohe offers a precedent for a more solitary, psychological condition. This relationship explores the body and an exterior that cannot be entered.

At the Farnsworth House, the glass envelope dissolves the boundary between inside and landscape. Mies said: "If you view nature through the glass walls of the Farnsworth House, it gains a more profound significance than if viewed from outside. That way more is said about nature — it becomes a part of a larger whole." The emphasis on nature through view is well represented in this project.

Another clear example of this is the unbuilt Resor House in Jackson Hole, Wyoming (1937–1938). This project was organized around a single panoramic view of the mountain landscape. Various collages related to this project show that the landscape was not incidental to the design but its primary subject, with the interior conceived as a platform from which to receive the exterior.



Fig. 17. Ludwig Mies van der Rohe, Resor House Project, Jackson Hole, Wyoming, c. 1937–41. Source: MoMA, 1941.

In a Martian context, this condition becomes necessary rather than aesthetic. The window is the only possible access to an uninhabitable exterior. This relationship between outside and inside is conceptualized well in Mies's work: while nothingness in space on Mars might seem like just an endless expanse of darkness and Martian soil, the significance it gains when carefully framed from the inside creates a psychologically powerful experience.

In this, framing becomes a psychological requirement. Architecture must return the outside world to the inhabitant through glass and aperture, holding the Martian landscape as a constant visual presence. The framed view becomes a tool for orientation, distance, and continuity with a world beyond the rigid interior.

II. Urban Frameworks for Collective Living

At the scale of urbanism, similar concerns appear in early modern attempts to reorganize cities not around image or monument, but around the organization of collective life, movement, and resources.

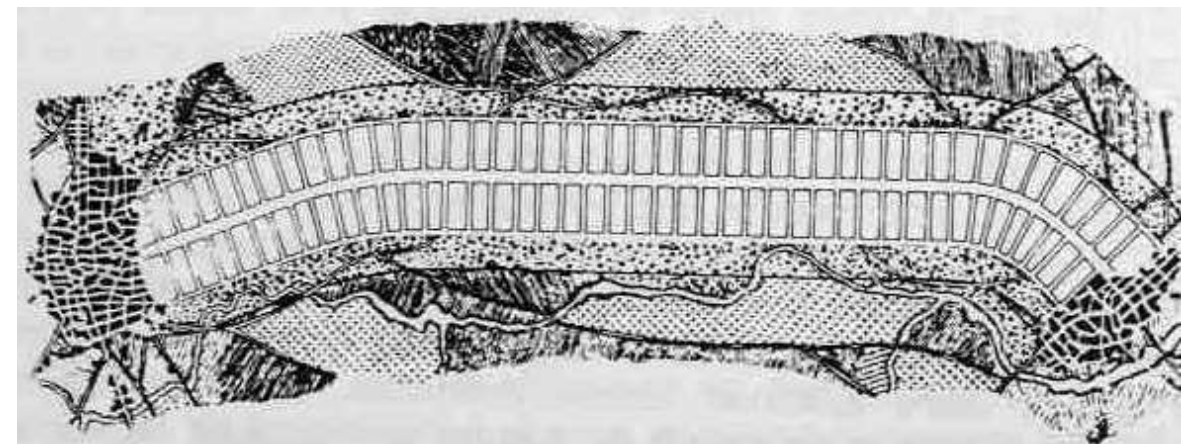


Fig. 18. Arturo Soria y Mata, concept plan for the linear city, 1882. Source: Arturo Soria y Mata, 1882.

Arturo Soria y Mata's Linear City proposes a layout where infrastructure, housing, and services are arranged along a continuous axis. Rather than reproducing the dense centers and peripheral edges of centralized models, the city becomes an extended field where access to systems is distributed along movement itself. Circulation is no longer secondary to urban form; it becomes the structural logic of the city.

A more radical formulation appears in Nikolai Milyutin's *Problems of Building Socialist Cities* (Sotsgorod). The city is structured around infrastructure, industry, and transportation rather than monuments or commercial centers. Though designed as a political model, its spatial logic is worth examining on its own terms. Sotsgorod minimizes separation between residence and production, so that living, working, and sustaining the settlement fold into a single continuous cycle.

Where Sotsgorod focuses on how urban form can organize collective life around shared infrastructure and resource efficiency, the present proposal reinterprets those principles through a more human-centered lens. Rather than treating inhabitants as components within a structured system, it explores how collective living on Mars might support social connection, individual agency, and long-term psychological well-being through the design of space.

Milyutin also understood communal institutions as essential social infrastructure rather than supplementary amenities. He positioned collective amenities at the center of everyday life, arguing that shared functions including dining facilities,

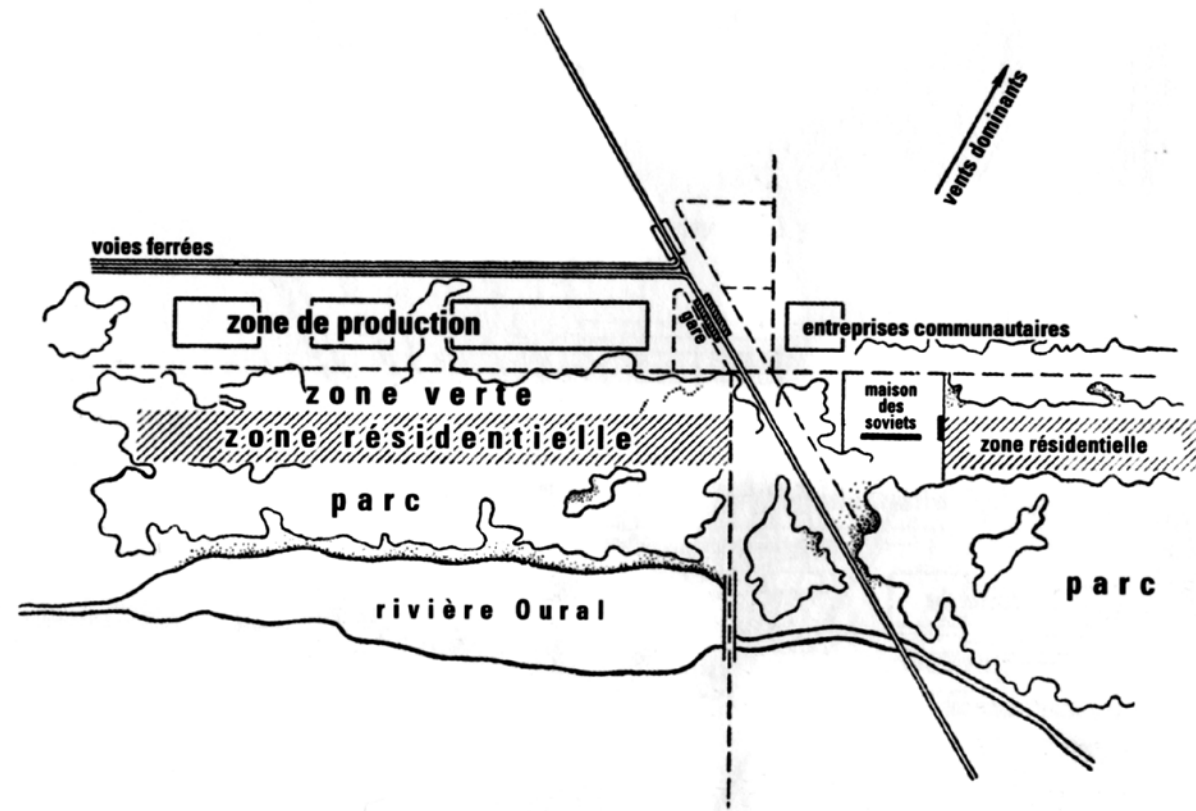


Fig. 19. Nikolai Milyutin, zoning diagram from *Sotsgorod: Problems of Building Socialist Cities*, 1930. Source: Nikolai Milyutin, 1930.

libraries, and recreation rooms should be connected directly to residential quarters and treated as the core of the settlement rather than its periphery (Miliutin, 1930/2021, Chapter 9).

Across both Linear City and Sotsgorod, urban form is understood not as image or object but as a spatial system of movement, access, and distribution. Circulation at the urban scale becomes an organizing principle for social structure itself. Rather than treating inhabitants as components within a structured system, it asks how collective living on Mars might support social connection, individual agency, and long-term psychological well-being through the design of space.

III. Habitation Beyond Earth

The final set of precedents extends architecture beyond Earth, where environmental constraint becomes the primary driver of spatial organization. Numerous competitions sponsored by NASA and large architecture firms have explored this territory, advancing strategies for sustainable extraterrestrial construction while raising broader questions about what kind of societies would form once humanity permanently settles.



Fig. 20. SEArch+ / Apis Cor, Mars X-House: Winner of NASA's Phase 3 3D-Printed Habitat Challenge in Final Design, rendering, 2018. Source: SEArch+, 2018.

NASA's 3D-Printed Habitat Challenge explored how autonomous construction and additive manufacturing could enable the rapid creation of habitats using Martian regolith. The first-place winner, Mars X-House by the SEArch+ / Apis Cor team (2018), demonstrated the technical feasibility of a 3D-printed Martian habitat through

integrated radiation shielding, in-situ resource utilization, and human-centered environmental design. Like many NASA-focused proposals, however, it operates at the scale of the building and mission rather than the settlement and society. Questions of social resilience, urban organization, and long-term adaptation remain

unresolved, suggesting an opportunity to investigate Mars not merely as a site for survival, but as a framework for human-centered communities capable of supporting meaningful collective life over generations.

Volume Zero's Marsception challenge pushed this conversation further by asking designers to consider how people might build communities, establish routines, and cultivate a sense of belonging in an isolated and resource-constrained world. The first-place winning project, *Thresholds* by Alec Naktin and Natalie Perri, proposes a hybrid Martian habitat embedded within a lava tube. Positioned at the interface between subterranean and surface environments, the design provides radiation protection while maintaining visual connection to the Martian landscape, addressing psychological well-being through natural light and openness. While the project valuably expands the conversation beyond technical survival toward lived experience, it remains focused at the scale of the individual dwelling. How multiple habitats connect, how communities grow, and how collective life is organized over time remain open questions.



Fig. 21. Alec Naktin and Natalie Perri, *Thresholds*, first-place winner, Marsception 2024 Architecture Competition, 2024. Source: Volume Zero Competitions, 2024.



Fig. 22. BIG (Bjarke Ingels Group), *Mars Science City*, Dubai, United Arab Emirates, 2018. Source: BIG, 2018.

significant gaps. It addresses survival and protection but does not engage with the social and psychological dimensions of settlement, including everyday rituals, loneliness, stress, or gradients of privacy, all of which are critical for long-term morale

and collective well-being.

Taken together, these precedents reveal a consistent gap across scales: from individual habitat to city prototype, the technical demands of extraterrestrial survival have been rigorously addressed, while the social, psychological, and urban frameworks necessary for permanent collective life remain largely unexplored.

IV. From Survival Systems to Social Systems

Across these precedents, architecture has been developed as a technical system than as a social one.

Aalto and Siza show that circulation and threshold organize habitation. Mies shows that the relationship between interior life and an inaccessible exterior must be actively designed. Soria y Mata and Milyutin show that movement and infrastructure, treated as primary structure, can shape collective life at urban scale. Together, they suggest that how people move, what they see, and how space sequences their encounters are not secondary concerns but they are the mechanisms through which architecture produces social life.

Yet all of these models still subordinate lived experience to systemic efficiency. The social remains a byproduct of the spatial.

The Mars precedents carry this tension to its limit. When survival depends entirely on the built envelope, technical demands overwhelm social ones. Mars X-House, *Thresholds*, and Mars Science City each solve a piece of the problem but none confront what it means to actually build and sustain a society in an extreme environment. On Mars, where every spatial condition is a careful and deliberate decision, the design of social life becomes inseparable from the design of survival. Architecture can no longer defer one in favor of the other.

Chapter 7

Methodology

I. Learning From Those Who Know

To understand what living in extreme confinement does to a person, I spoke with people who have experienced or studied it directly. I interviewed Jay Apt, a former NASA astronaut who flew on four low-Earth-orbit missions; Georgi Petrov, Chair of the Space Architecture Society; and James Wynn and Kathy Newman from Carnegie Mellon University, whose work explores speculative futures through a humanities lens. These conversations did not provide definitive answers. Instead, they directed me toward the right questions driving this thesis.

When I asked how he and his crew coped with isolation, Jay Apt described how astronauts spent their free time in orbit. “Some people would put on music on their headphones and use free time to look out the window... to get away from it all with their music. I like looking out the window and taking pictures,” he said. In an environment of total constraint, the window is not a luxury but an essential instrument of orientation. It provides a point of reference within the immensity of space, allowing astronauts to situate themselves within a larger whole.



Fig. 24. Astronauts gathered for a meal aboard the International Space Station. Source: NASA.



Fig. 23. Dr. Jay Apt using a Hasselblad camera aboard the Space Shuttle Endeavour during STS-59, 1994. Source: NASA, 1994.

Apt was equally direct about the importance of ritual. “Eating together matters. It is one of the few daily rituals that creates normalcy, reinforces crew bonds, and supports morale in confined environments.” Under extreme conditions, routine becomes spatial as much as social.

Shared meals are not merely moments of nourishment; they are rituals that structure daily life, strengthen interpersonal relationships, and support psychological well-being.

Apt also described how environmental qualities like light could transform the experience of confinement. “When we docked with Mir, the outside was covered in white beta cloth. It was brilliant white. Having this brilliant white thing outside your overhead windows, it was like being outside on a sunny day.” Even in highly artificial environments, sensory cues remain essential. Light, color, and material qualities reconnect inhabitants to familiar earthly experiences, mitigating the psychological effects of isolation and enclosure.



Fig. 25. Space Shuttle Atlantis docked with the Mir Space Station during STS-71, 1995. Source: NASA, 1995.

While these observations offer valuable insights into life in confined environments, they come primarily from low Earth orbit missions, where astronauts remain closely connected to Earth through communication, visible planetary context, and the possibility of return. A Mars settlement introduces fundamentally different conditions of distance, autonomy, and permanence. The challenge therefore shifts toward establishing long-term belonging, identity, and home.

Across these conversations, however, one idea remained constant: **certain human needs persist regardless of environment.** Social connection, movement, nourishment, ritual, and a sense of belonging are not secondary conditions but essential to survival. **Architecture, in this sense, becomes not only shelter, but a framework that supports these conditions.**

Quotes from Former NASA Astronaut Jay Apt



Fig. 26. (Left) Zoom interview with former NASA astronaut Jay Apt conducted by the author, 2026.

Fig. 27. (Right) Jerome Apt, mission specialist on STS-79, 1995. Source: NASA, 1995.

“Some people would put on music on their headphones and use free time to **look out the window...** to **get away from it all** with their music. I like looking out the window and taking pictures.”

“Eating together matters. It is one of the few daily rituals that **creates normalcy,** **reinforces crew bonds,** and **supports morale** in confined environments.”

“When we docked with Mir, the outside was covered in white beta cloth. It was brilliant white. Having this **brilliant white** thing outside your overhead windows, it was like **being outside on a sunny day.** Really different from just having space out there.”

“Windows are always good. **Mars is pretty desolate,** but so is the **desert southwest.** People live in Arizona with big picture windows. So I think that would be the one feature.”

II. Social Connection

The need for spaces that support connection, movement, nourishment, ritual, and a sense of belonging is reinforced by decades of psychological and behavioral research. One of the most influential examples is the Harvard Study of Adult Development, often referred to as the Harvard Longevity Study (Waldinger & Schulz, 2023).

Beginning in 1938, the study followed approximately 2,000 participants across four generations, tracking physical health, mental health, careers, relationships, family life, and overall well-being over the course of their lives. Designed to answer the question “What makes a good life?”, it has become one of the longest-running longitudinal studies of human development.

Across nearly nine decades of observation, researchers found that the strongest predictor of health, happiness, and longevity was not wealth, fame, or social class, but the quality of a person’s relationships. **Participants with strong social connections tended to live longer, experience better physical and mental health, and report higher life satisfaction.** Loneliness and isolation, by contrast, were consistently associated with poorer health outcomes. Importantly, the study also

“Good relationships don’t just protect our bodies; **they protect our brains.**”
- Robert Waldinger

found that relationship quality mattered more than quantity: a small number of close, supportive relationships was more valuable than a large social network.

These findings suggest that the environments in which people live should be evaluated by their capacity to support social connection. If strong relationships are among the greatest predictors of health and longevity, then spatial design becomes more than shelter. It becomes a framework through which social life is formed and sustained.

For a thesis set on Mars, this becomes particularly urgent. **In a constrained environment where social circles are limited and privacy is scarce, opportunities for interaction are mediated almost entirely by architecture.** Psychological well-



Fig. 28. Circulation becomes shared space. Community Design Agency, Sanjaynagar Slum Redevelopment Project, India, n.d. Source: Rajesh Vor.

being cannot be separated from spatial design.

Viewed through this lens, Mars becomes a testing ground for a larger question: if social connection is fundamental to human flourishing, how might environments be designed to actively support it?

III. Spatial Isolation in Constrained Environments

While the Harvard Study of Adult Development assumes Earth-based conditions where access to others and environmental variability are relatively stable, Mars removes these assumptions. It is therefore necessary to examine the psychological and behavioral consequences of sustained isolation in constrained environments.

One of the foundational references is the Mars500 experiment conducted by the Institute of Biomedical Problems (IBMP) in collaboration with the European Space Agency (ESA). The study simulated a full-duration Mars mission (520 days) within a closed habitat, with delayed communication and restricted environmental variation.



Fig. 29. Mars500 “spaceship” isolation simulation crew module, Moscow, Russia, 2009. Circulation becomes shared space. Source: ESA – S. Corvaja.

The study found that there were major stress factors that included monotony of environment, loss of time awareness, and social friction within small groups (Mars500 Study, 2011). While the performance remained generally the same, there were heavy fluctuations in mood, motivation, and sleep quality. This suggested that environmental factors strongly influenced privacy

needs, conflict frequency, and need for psychological recovery period.

Taken together, these studies point toward a design question as much as a psychological one: built environments actively construct the conditions for relational life under constraint. In extreme environments, well-being depends on spatial organization at both the societal and intimate scale. The ability to move between gradients of privacy, mediating exposure and withdrawal, becomes essential for maintaining social cohesion and psychological stability.

IV. Technical Research

Radiation is one of the primary environmental constraints shaping habitation beyond Earth. Mars has no global magnetic field and an atmosphere less than one percent the density of Earth's, while the Moon lacks both entirely. Without these protective filters, the surface is continuously exposed to galactic cosmic rays, solar energetic particles, and radiation (Cucinotta et al., 2008; Zeitlin et al., 2013). In such environments, shielding is not an added layer of protection but a fundamental architectural requirement.

One response is inflatable structural systems. Unlike rigid modules constrained by launch volume, inflatable habitats can be transported compactly and expanded on site. Early demonstrations such as Bigelow Aerospace's BEAM module showed that expandable systems can maintain stable pressurized environments in orbit (NASA,

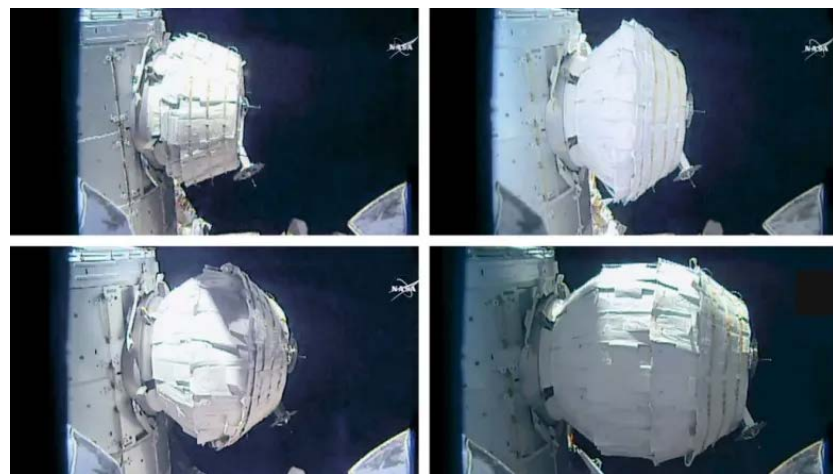


Fig. 30. A series of photos tracks the expansion of the BEAM module on the ISS May 28. Credit: NASA

2016). These systems rely on layered wall assemblies composed of pressure bladders, structural restraint fabrics, and protective shielding materials. Hydrogen-rich substances such as polyethylene are effective in reducing radiation exposure,

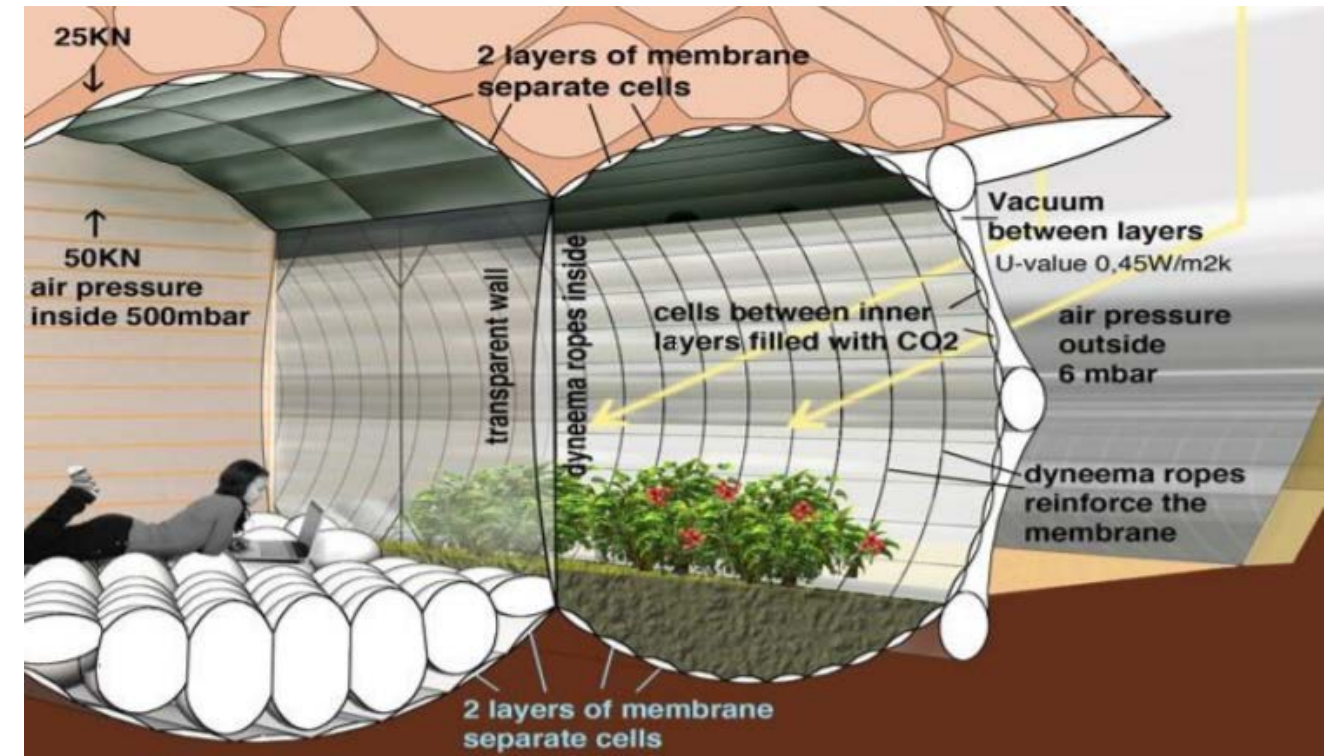


Fig. 31. Space Nomad, section through wall and roof structure. Source: Gábor Bihari and Thomas Herzig.

and water can serve both as shielding and life-support resources (Durante & Cucinotta, 2011). The habitat envelope becomes an integrated environmental system rather than a simple boundary.

However, material shielding alone is insufficient. The required mass exceeds practical launch limits, shifting the focus toward site-based protection strategies. Martian regolith and crater formations provide naturally available shielding material that can be used to embed or cover habitats (Simonsen et al., 1990; Hofmann et al., 2020). Subsurface placement further reduces thermal variation and radiation exposure.

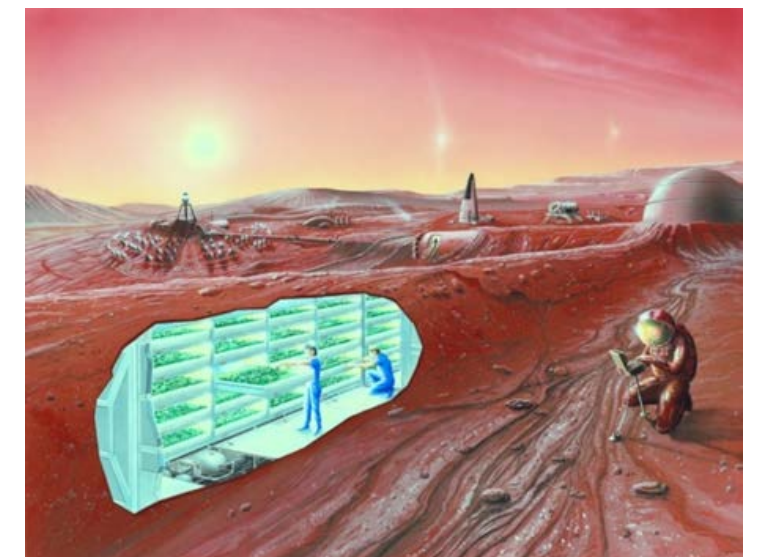


Fig. 32. NASA concept of underground farming on Mars. Source: NASA.

This condition suggests a potential hybrid configuration in which an inflatable habitat is nested within an excavated landscape. The crater wall functions as an outer protective shell, while the

membrane provides a pressurized and habitable interior. Neither system is sufficient independently; together they create a layered form of protection in which the membrane filters and the landscape absorbs (Hofmann et al., 2020).

The resulting geometry is further informed by structural behavior. Internal air pressure exerts force equally in every direction, making curved forms inherently efficient. Flat surfaces concentrate stress at edges and corners, whereas continuous curvature distributes loads evenly across the structure (Wagner et al., 2000). Spheres and cylinders therefore emerge not as aesthetic choices but as geometries dictated by pressure itself.

Reduced gravity reinforces this logic. Mars experiences only thirty-eight percent of Earth's gravity, and the Moon approximately sixteen percent. Lower gravitational forces reduce the compressive loads that typically govern terrestrial construction, allowing lighter structures to perform effectively where heavier ones would be impractical. These factors can be understood as converging constraints that inform architectural response.

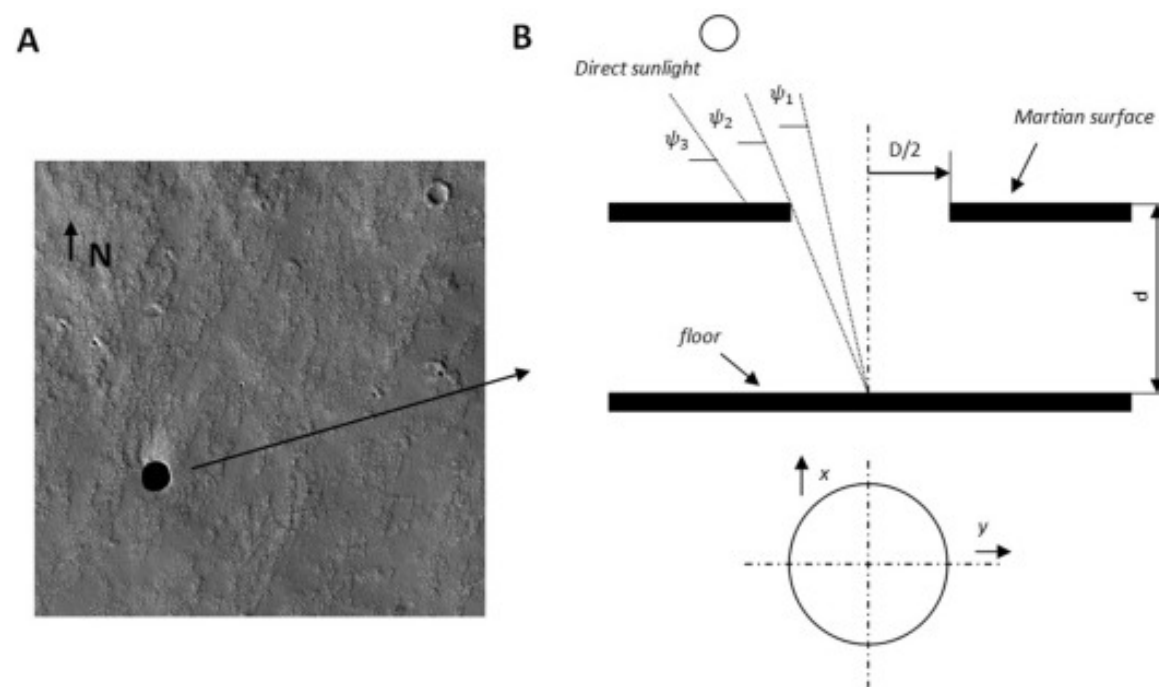


Fig. 33. (left) Aerial view of northeast of Arsia Mons, Mars, where cavernous spaces are expected. Credit: MRO/HiRISE.

Fig. 34. (right) Schematic of a modeled void illustrating how sun elevation angles govern whether direct sunlight penetrates to the void floor or leaves it in shadow. The reference system is centered on the void and normalized to its aperture. Source: D. Viúdez-Moreiras.

Chapter 8

From Research to Design

Taken together, the interviews, behavioral studies, technical investigations, and architectural precedents suggest that habitation in extreme environments is fundamentally a question of organization rather than enclosure.

The precedents demonstrate this across scales. Aalto's Paimio Sanatorium shows how architecture can support well-being through carefully calibrated relationships between movement, light, nature, and social life. Siza's housing projects position circulation as a social framework, creating opportunities for encounter through shared thresholds and collective passage. At the urban scale, Linear City and Sotsgorod organize settlement around infrastructure, movement, and collective resources rather than isolated objects. Extraterrestrial habitat proposals similarly reveal that survival depends upon integrated systems of resource production, environmental control, and protection.

The research reinforces these observations. Astronaut interviews highlighted the importance of orientation, ritual, and sensory connection to the outside world. Studies of social well-being emphasized the central role of interpersonal relationships, while confinement research identified monotony, social friction, and loss of temporal structure as recurring challenges in isolated environments. Technical investigations demonstrated that environmental systems, resource cycles, and protective envelopes become inseparable from architecture itself.

Collectively, these findings suggest several spatial implications. Circulation operates as more than movement; it becomes a framework for encounter and orientation. Collective spaces function as social infrastructure, supporting rituals that structure everyday life. Privacy is most effective when organized as a gradient rather than a fixed separation, allowing inhabitants to move between solitude and community. Environmental variation, views, and natural rhythms become essential tools for psychological well-being, while infrastructure shifts from a hidden system to an active component of habitation.

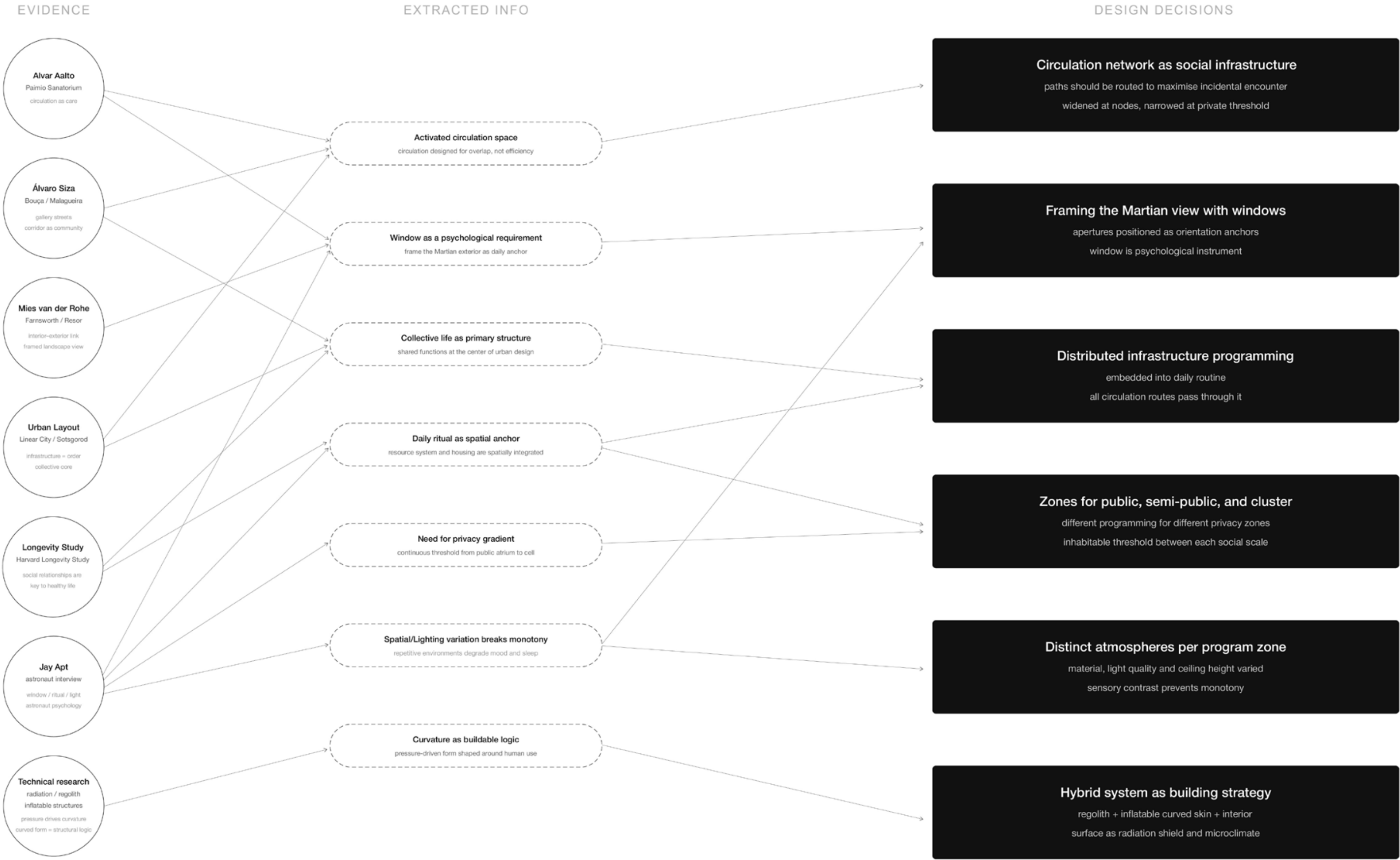


Fig. 35. Diagram illustrating the translation of precedent design evidence into informed design decisions. Source: Author.

Chapter 9 In 2500.



The settlement has been here for generations. Long enough that no one remembers arrival, only the rhythms that followed. The routines of tending, repairing, gathering. The knowledge that every resource is shared, every action connected to someone else's survival.

Mars remains unforgiving. But life here is no longer defined by endurance alone. What began as a challenge of survival has become a question of how to live together.

The Site

Ice, Minerals, and Infrastructure

The project is situated in Utopia Planitia, in the year 2500, where subsurface ice and minerals at the bottom of the crater become the primary resource. First identified by NASA in 2016, the deposit is no longer a geological discovery but essential infrastructure. It provides water, oxygen through electrolysis, and hydrogen for energy production, allowing the settlement to operate with a high degree of independence.

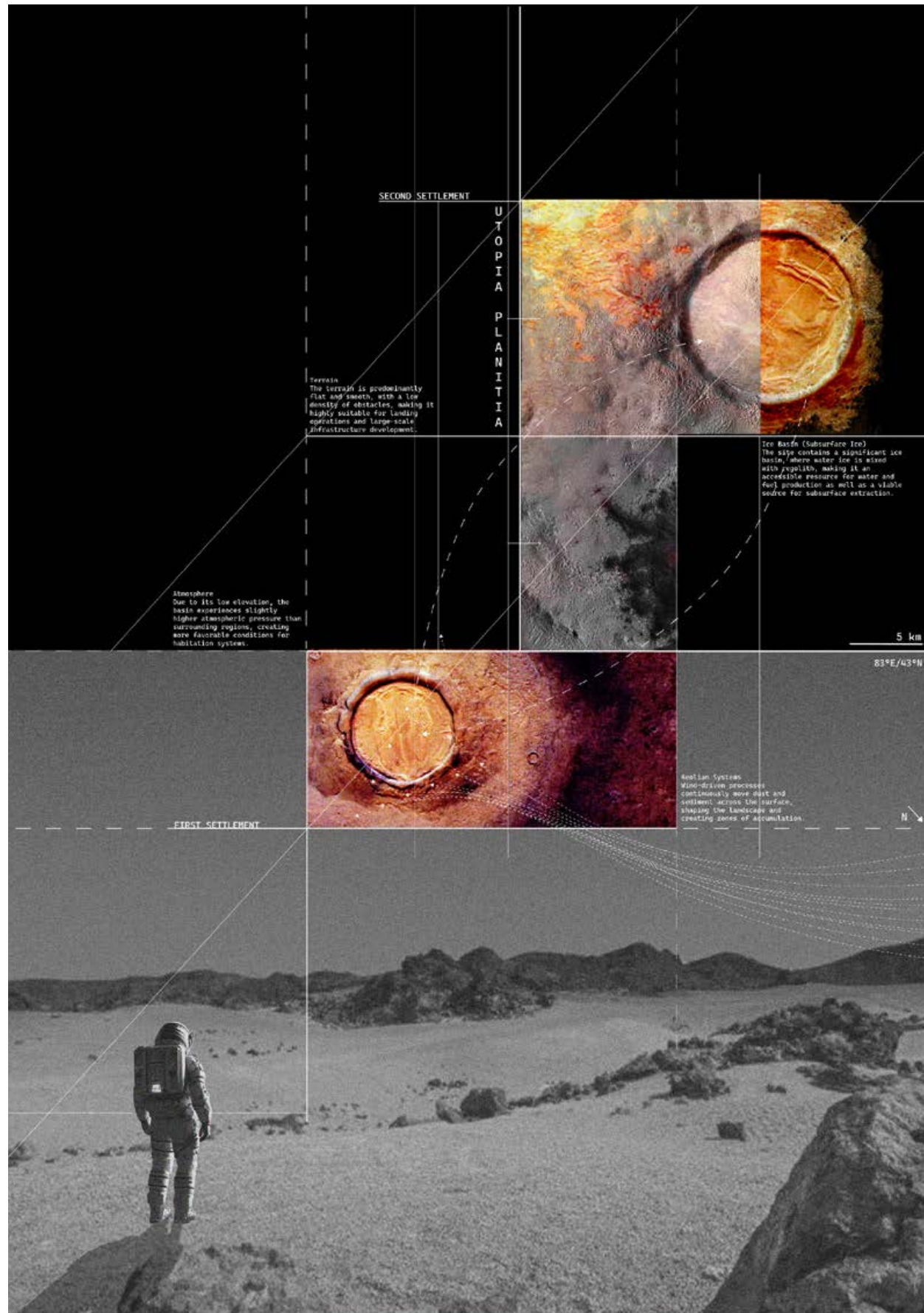
The crater is equally important. Its walls and surrounding regolith provide natural radiation shielding and thermal stability, reducing the need for additional protective construction. At the same time, the crater creates variation in elevation, light, and views, offering a more legible environment than the surrounding plains. These qualities support orientation, navigation, and psychological well-being over long periods of habitation.

Settlement Growth

The settlement develops gradually. The first phase occupies the rim of a five-mile crater, where level ground allows for early construction. As the settlement grows, development moves inward and downward, replacing temporary surface structures with permanent buried environments that take advantage of the crater's natural protection.

A second crater accommodates future expansion. The two settlements are connected by a dual-band linear structure adapted from Georgi Petrov's redundant settlement model. Where the bands converge, shared facilities and civic spaces emerge, allowing the settlement to densify without restructuring its overall organization.

Fig. 37. Site plan depicting settlement growth strategy within Utopia Planitia, Mars. Source: Author.



Infrastructure First

The first settlement cuts across the crater along a single connective spine. At the left, the lowest point, sits the ice mining and infrastructure hub, where the richest deposits of subsurface ice and minerals are concentrated. From here the network extends outward, processing systems feed into a greenhouse and communal kitchen, which connect onward to housing, a gym, and a meditation space anchored at the far right. Geography drives this arrangement, and the reasoning will be examined more closely later.

Rather than a single centralized system, the settlement develops as a network of ice mining hubs, processing systems, and transportation nodes spread across the crater floor. Each operates with enough independence to keep the whole resilient. Bio-regenerative nodes distributed throughout cycle water, air, and nutrients locally, so no single failure brings the system down.

Ice extracted from the floor is processed into water, oxygen, and hydrogen. Mineral deposits yield construction material and agricultural substrate. The crater is both the source of the settlement's resources and the framework that shapes its growth, the settlement doesn't simply occupy it, it is shaped by it.

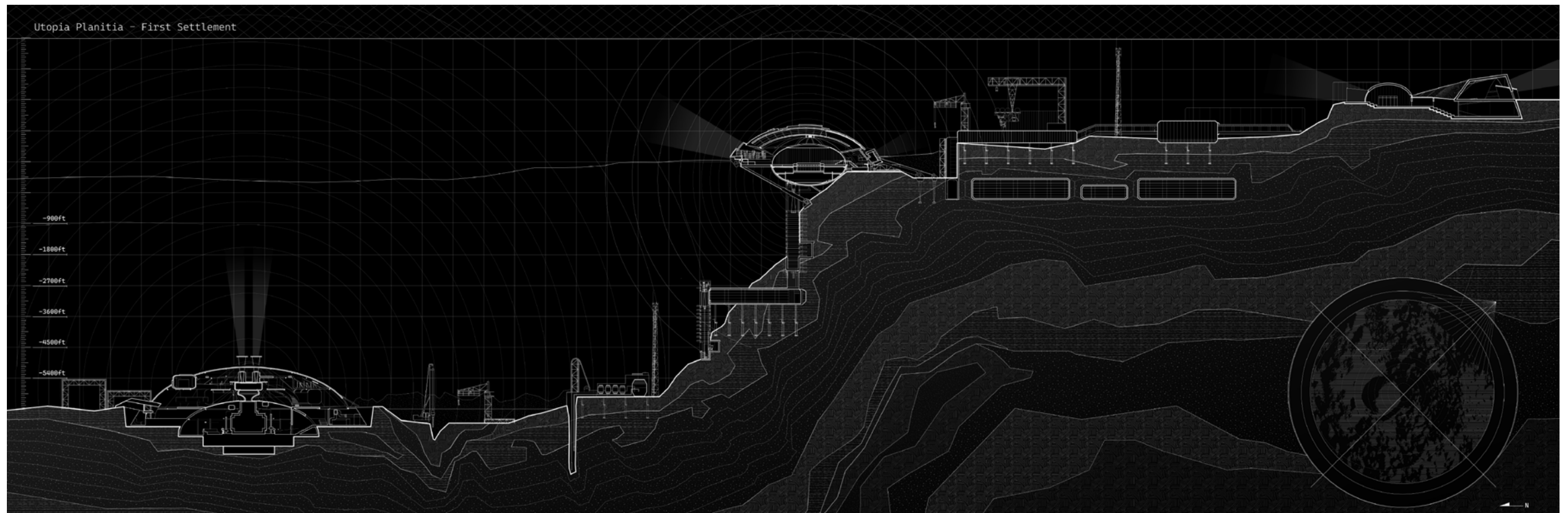
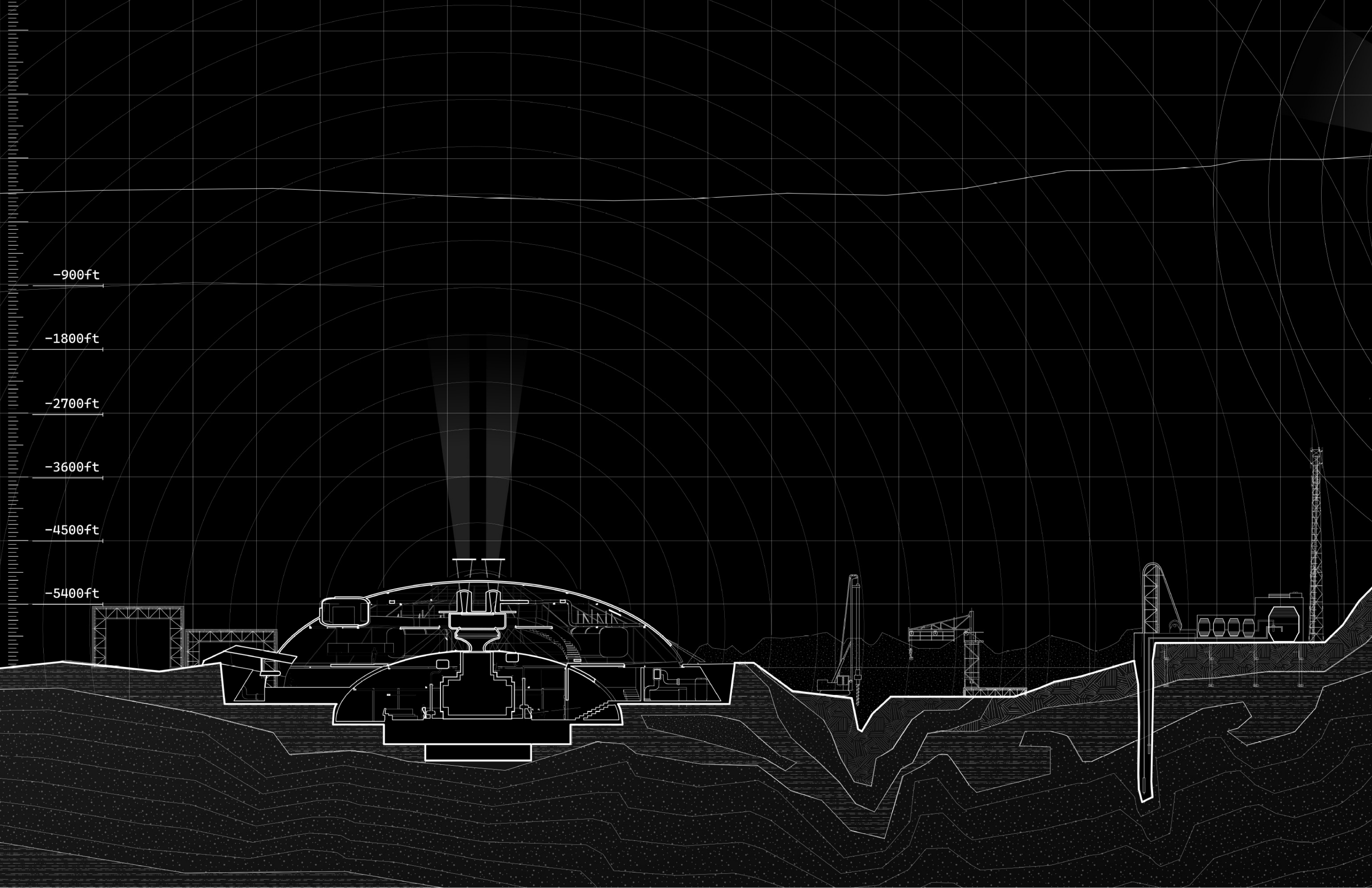


Fig. 38. Section through the first settlement of Utopia Planitia, 2500. Source: Author.



-900ft

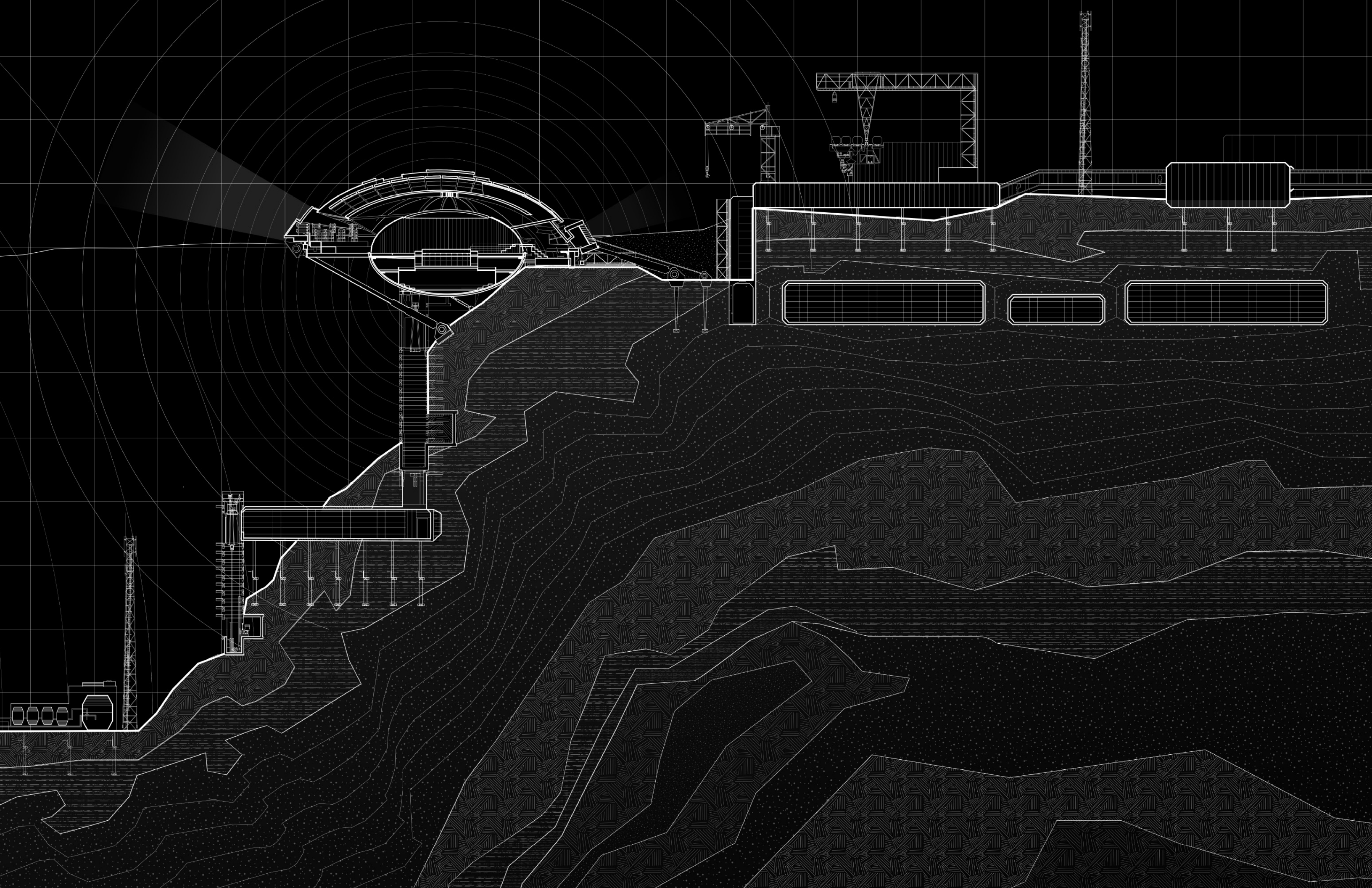
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-2700ft

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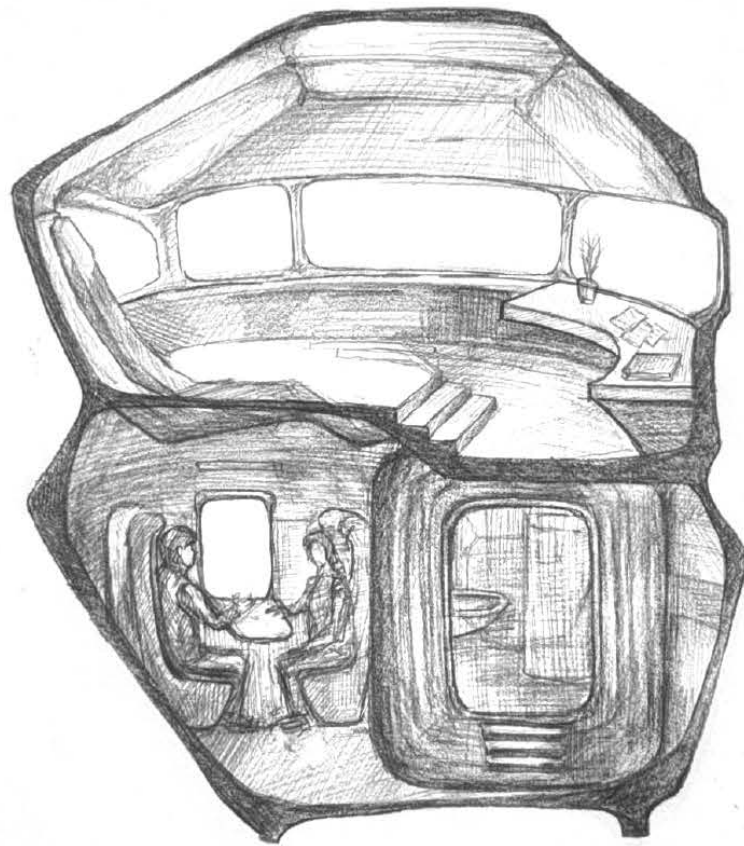
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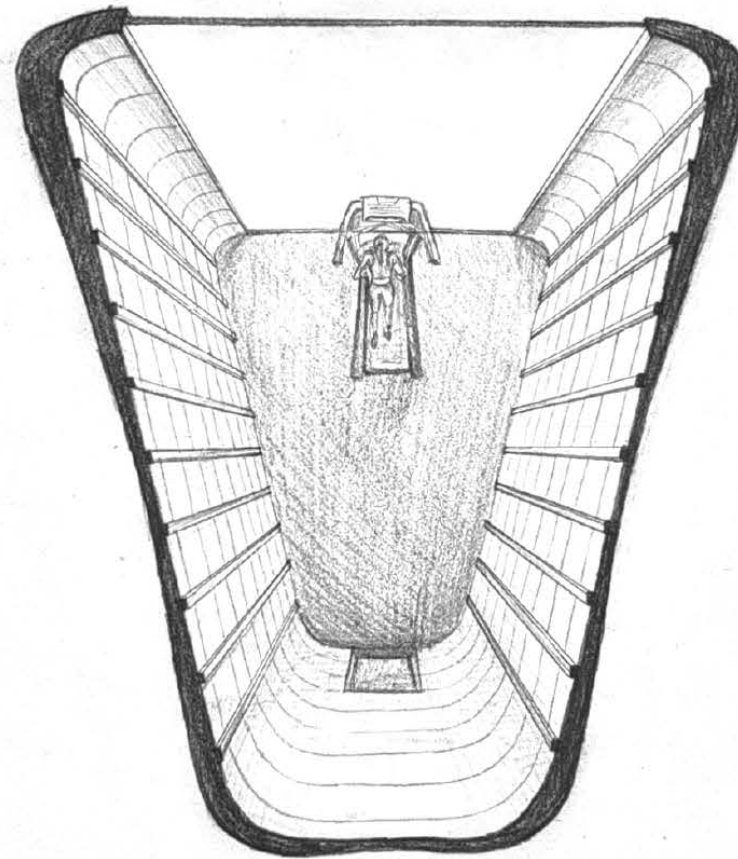


The Early Iterations: Sketches

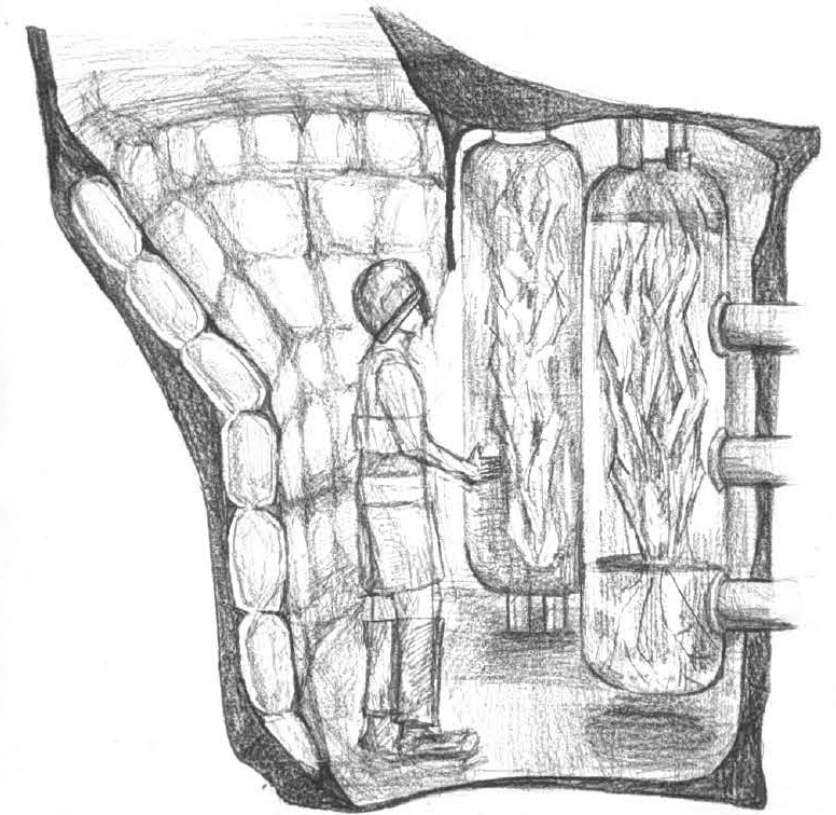
The early sketches trace first question. What does it mean to be human here?



The living quarter asks how people find each other, gather, and rest.



The exercise room asks how the body stays functional, and what it means to frame a view outward when outside is uninhabitable.



The greenhouse asks how a closed system feeds itself, and what happens when nature and infrastructure become one.

Fig. 39. Hand Sketches of the living quarter, exercise room, and greenhouse. Source: Author.

The Early Iterations: Underground City

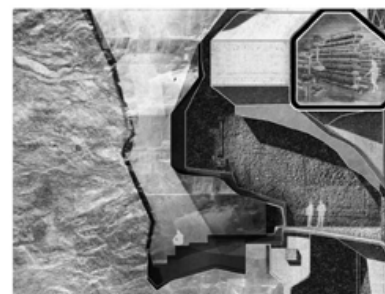
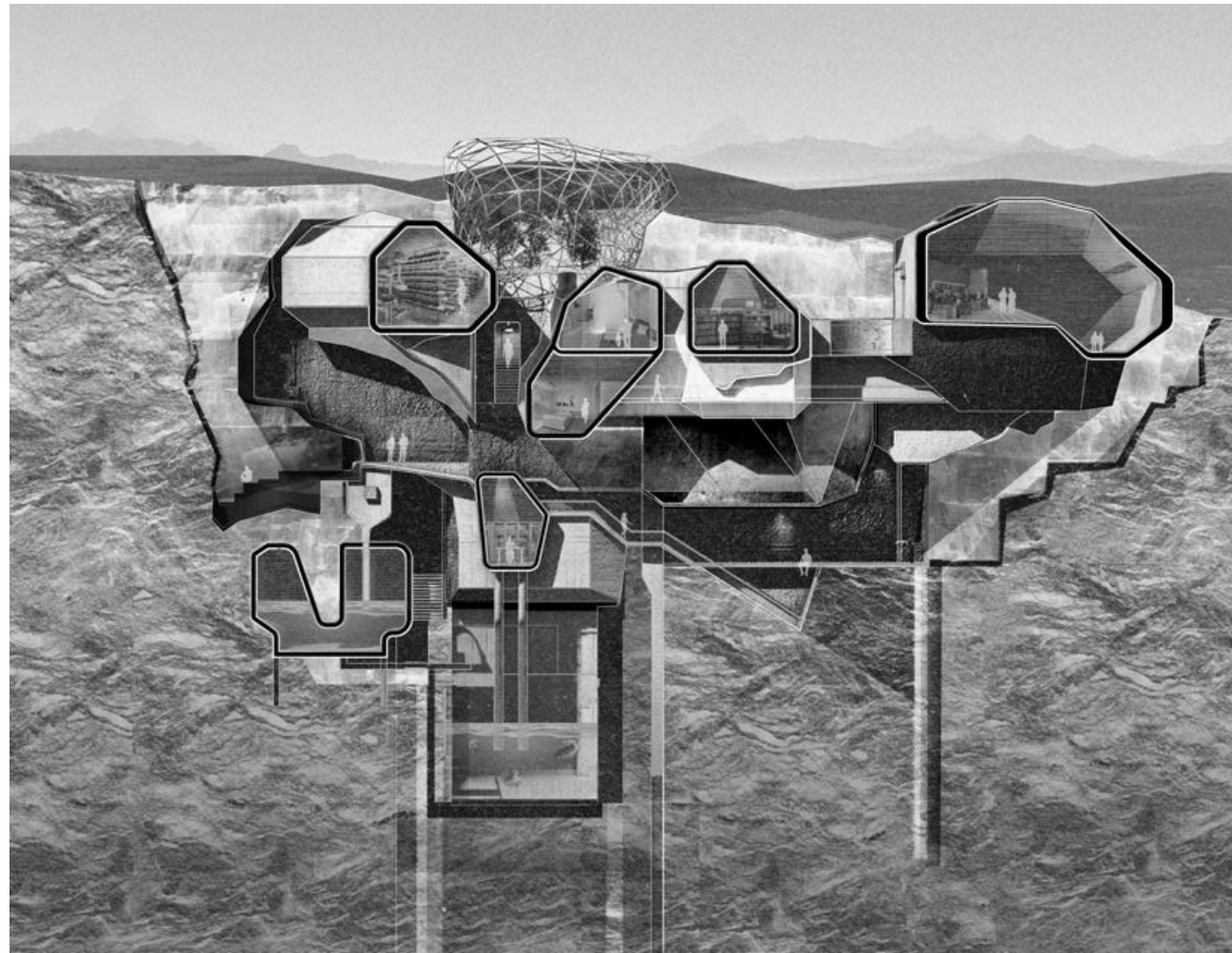
One answer to those questions was to go underground entirely. The project operates through thick, embedded, excavated volumes, responding to radiation, temperature extremes, and the absence of any protective atmosphere. This is the logic Rudofsky documented in *Architecture Without Architects*: settlements pressed into hillsides, dug into earth, shaped by necessity rather than profession.

Carving generates depth, gradients of light, interiors that feel continuous rather than contained.

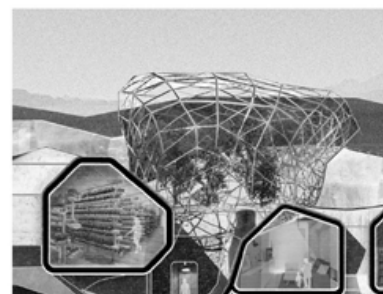
Within this, scale starts to differentiate experience.

There are spaces where the body feels small: the light tunnel, the communal dining room, the circulation hub. And spaces where the body feels held: the hydroponic gardens, the housing.

But this direction had limits. The further the architecture retreated underground, the darker it became. Light had to be fought for, routed, rationed. And radiation did not disappear below the surface. It followed, accumulated, became harder to control. Going deeper was not the same as going safer. The next move would have to find a different relationship between protection and light.



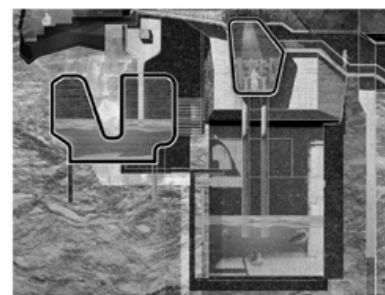
the light tunnel



hydroponic gardens / greenhouse



communal dining halls / observatories



ice mining storage / water control tower



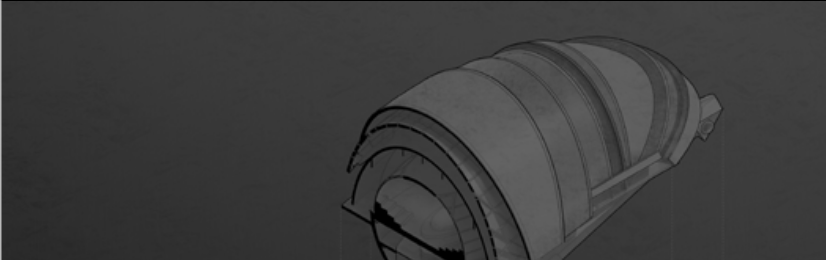
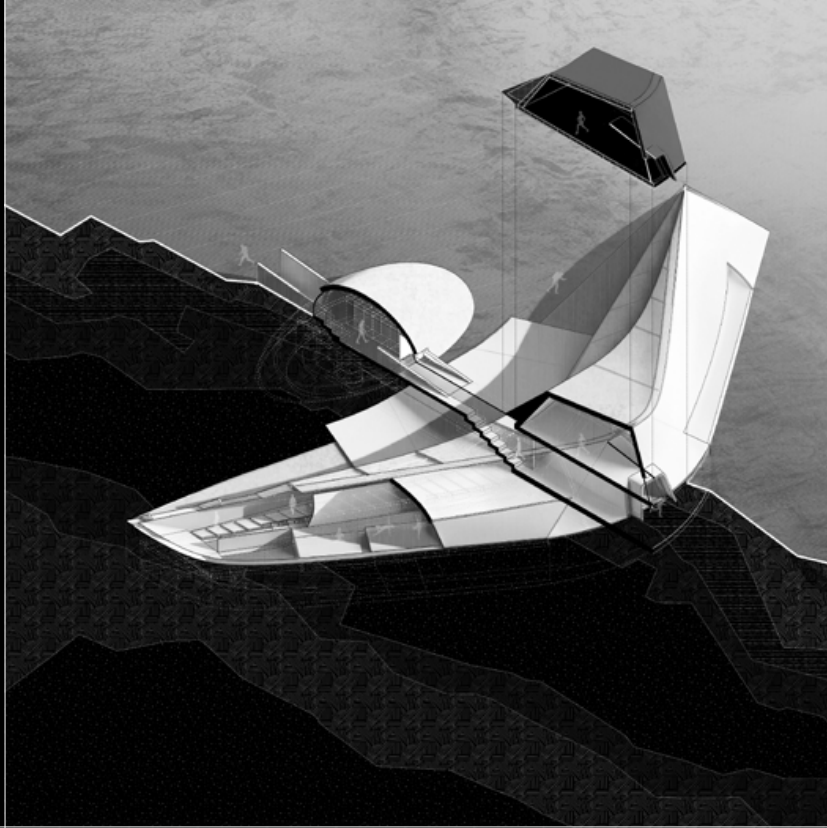
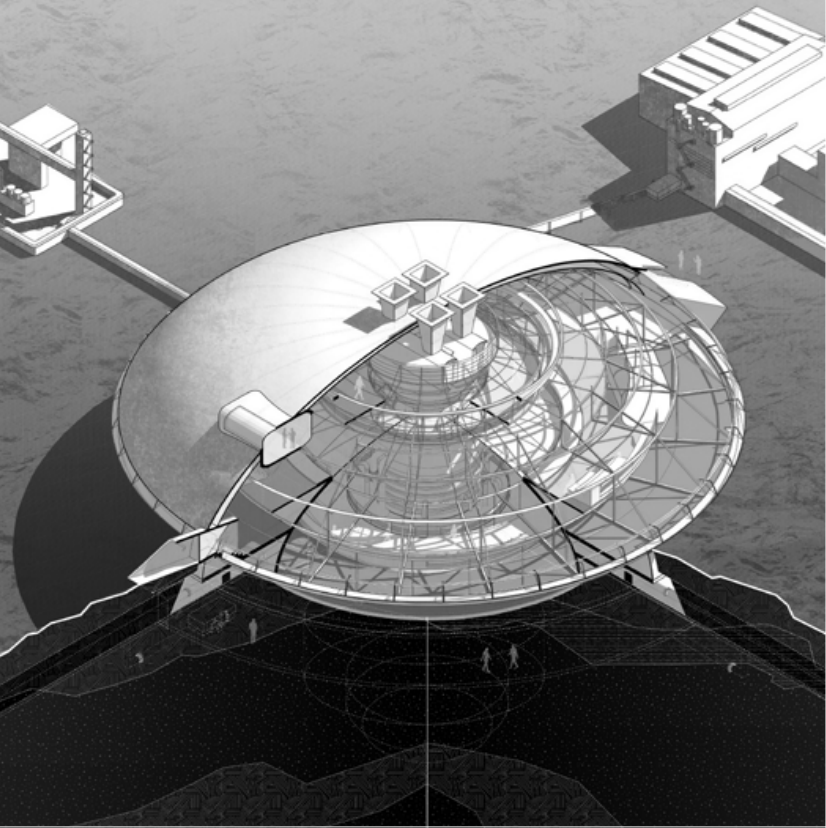
(left) collective housing / (right) individual housing



circulation hub

Fig. 40. Underground Martian City. Source: Author.

Final Design



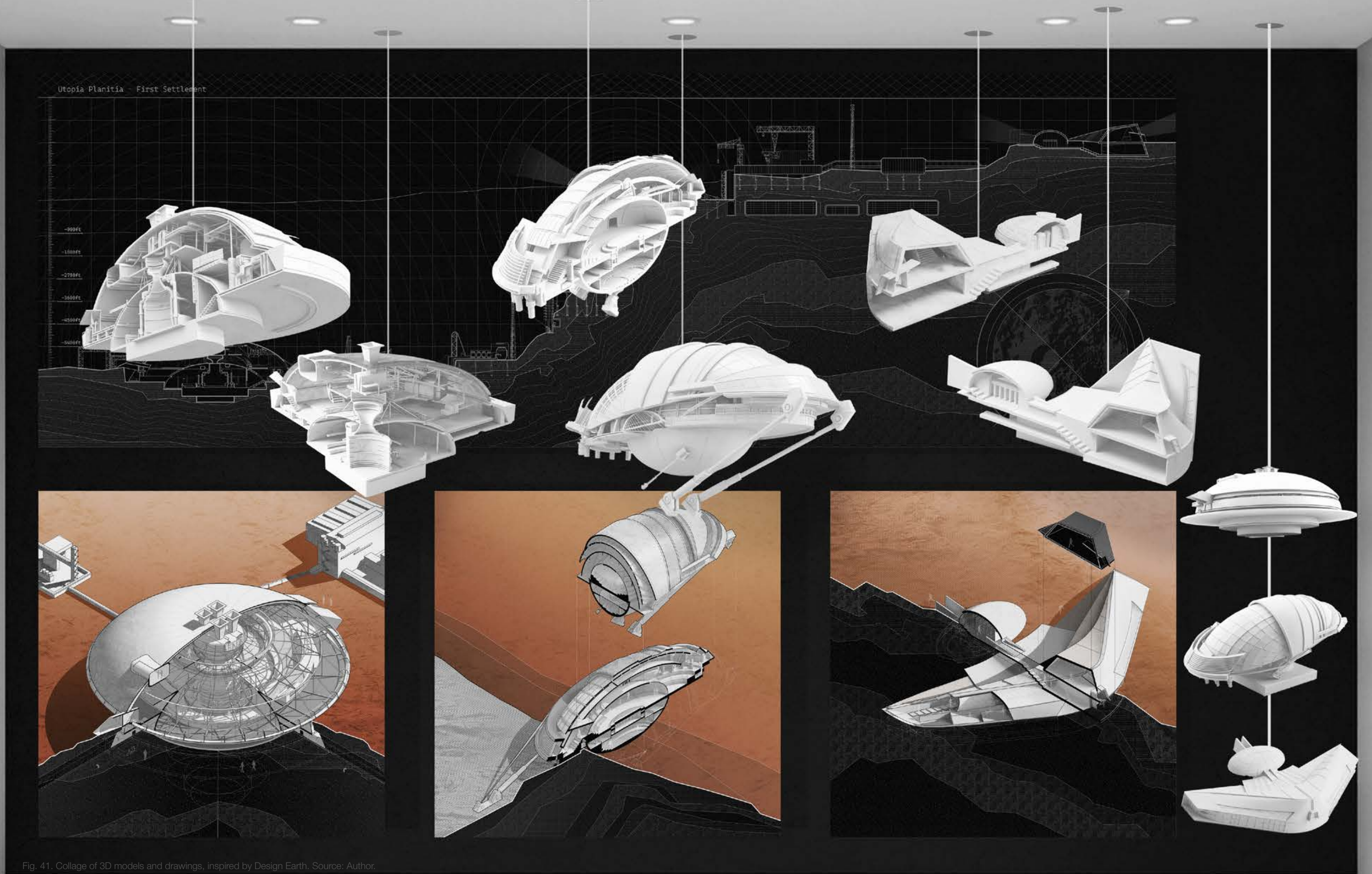


Fig. 41. Collage of 3D models and drawings, inspired by Design Earth. Source: Author.

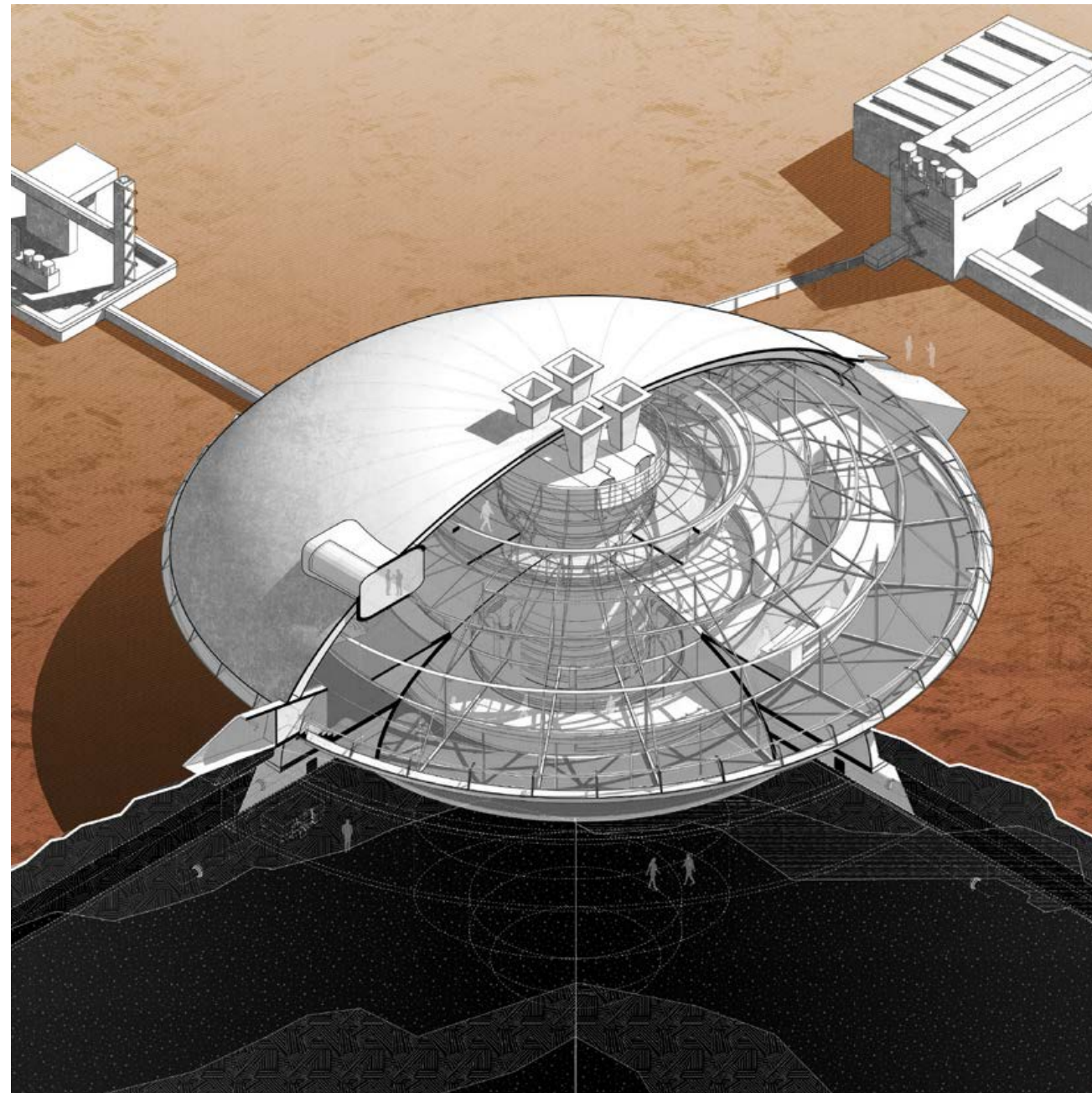
The Cycle: The Infrastructure Hub

This is the largest shared space in the settlement. It is where water is processed, air is cycled, waste is returned to the system, and extracted ice is processed and returned to the inhabitants as a resource. It is not a civic building with infrastructure hidden underneath it. It is the infrastructure, and people live inside it.

You enter from below, through a corridor that narrows before it opens into the full volume of the hub. The plan is circular. **Circulation follows the curve, which means you are always moving past someone, always crossing paths.** Along the perimeter, small pods break off from the main flow, places where people can stop, sit, and gather, directly against the exposed infrastructure. Nothing is hidden behind service panels.

The hub operates across multiple layers. Some run underneath the ground, carrying the deeper mechanical processes down and out of reach. Some rise above, stacking function on top of function, each level legible from the ones below. People move between them, or they stop. They settle into the capsule cubicles set into the walls of each layer and they watch it all happen around them. They talk. And as they do, the infrastructure recedes, not because it disappears, but because it becomes familiar. It becomes the background.

The systems that keep you alive are just there. **Not hidden, not celebrated. Present, constant, and ordinary in the way that only essential things can be.**



Water, air, waste. Everything cycles back. People too, at the end of a long shift, finding each other here. What keeps them alive is also where they gather. The pipes, the levels, the hum. Not hidden. Just home.

Fig. 42. Section isometric of the infrastructure hub. Source: Author.

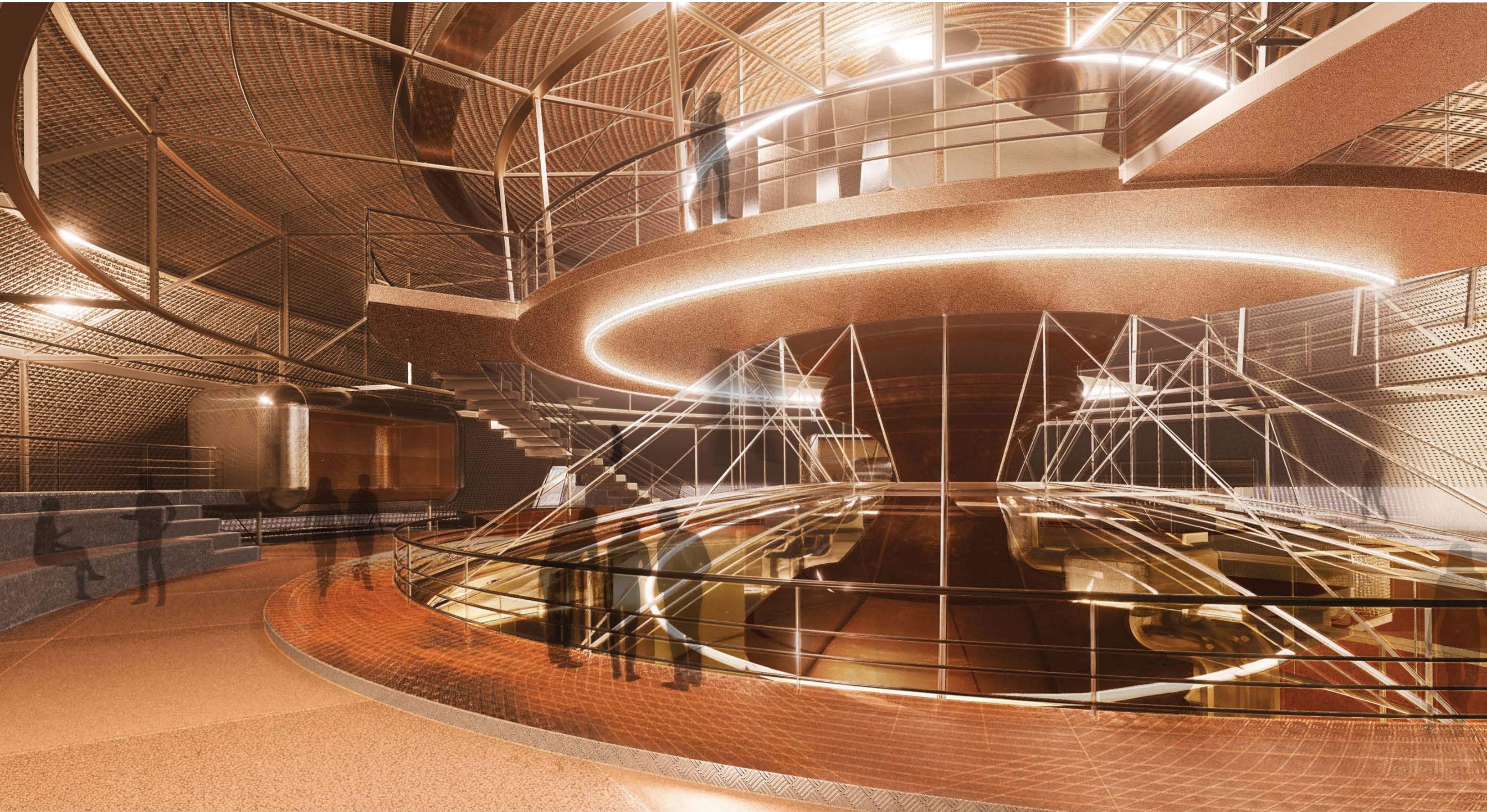


Fig. 43. Interior rendering of the infrastructure hub, featuring communal gathering, and a central ice-water energy generator. Source: Author.

The Tending: Greenhouse and Home

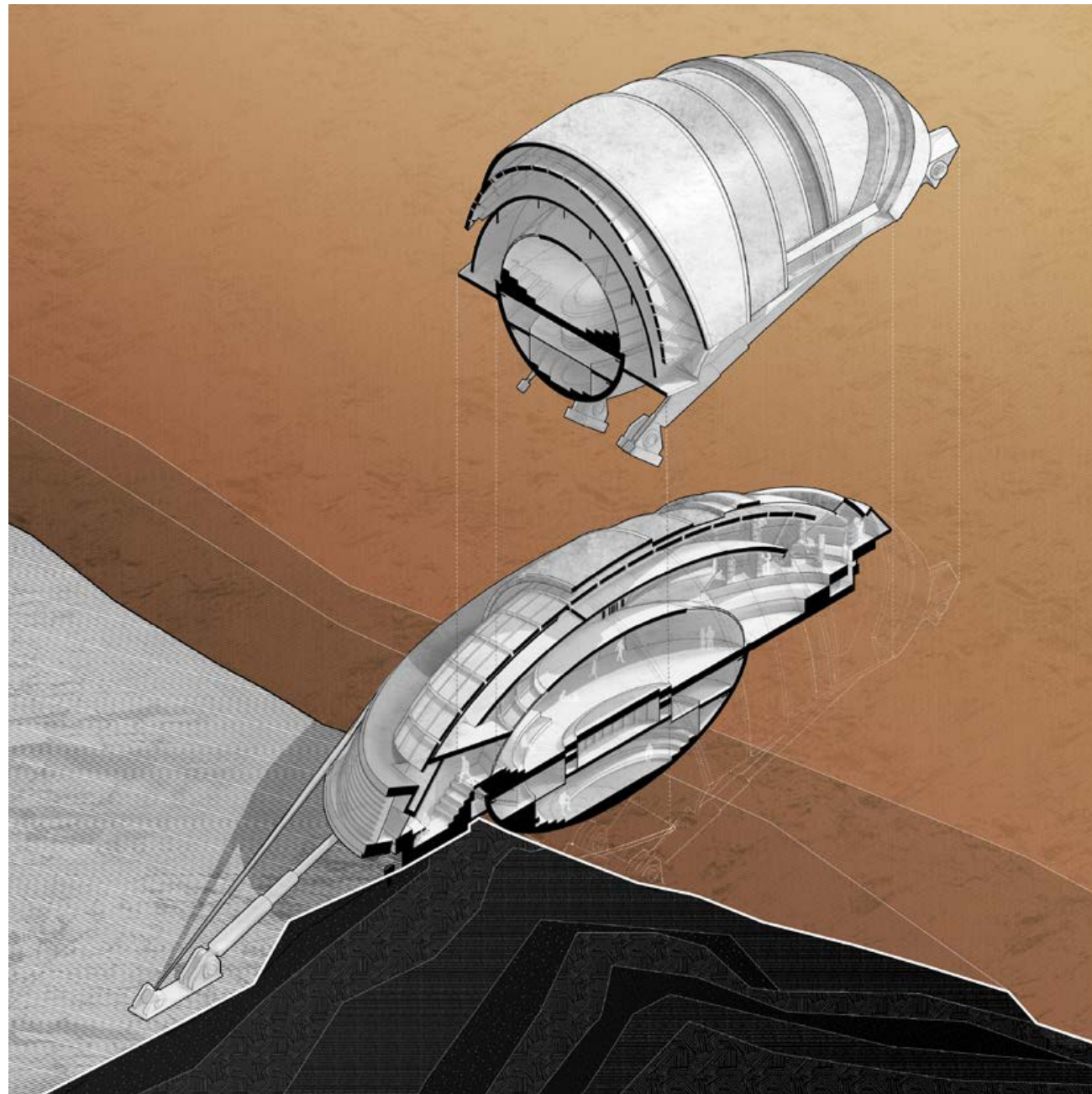
This is where the settlement grows its food, cycles its oxygen, and reclaims its water. It is also where people eat together at the end of the day.

The greenhouse sits on a cliff edge, positioned for maximum light exposure from the surface and oriented to look back toward the living quarters. Light enters diffused, filtered through hydrogen panels that scatter it evenly across the growing beds, softening the harshness of the Martian sun into something closer to dusk on Earth. Deeper in, controlled-spectrum strips supplement what the surface cannot provide, tuned to the needs of each crop and to the rhythms of the people tending them.

The path through it winds around the growing beds. It does not direct you. It gives you somewhere to be. Over time, the walk becomes a ritual, a daily passage through the living infrastructure of the settlement, a moment when the mechanics of survival are held close enough to touch. On Mars, where nothing sustains itself without intention, this proximity is not incidental. It is how a culture learns to understand what keeps it alive.

At the far end, the path opens into the communal dining space. Beyond the windows, the living quarters look back across the cliff. The food on the table comes from the room you just walked through. At the end of the journey, there is a home, not a facility, but a place where people sit down together and eat what they grew.

The social and the ecological are the same loop. Infrastructure does not sit beneath life here. It is woven into the texture of the day.



Every leaf accounted for. Every root a small miracle. Nothing grows here by accident. She comes here the same time each day, the way people have always returned to the things that keep them alive.

Fig. 44. Section isometric of the greenhouse/home. Source: Author.



Fig. 45. Rendering of greenhouse. Source: Author.

The Pause: Exercise Room

On Mars, muscle atrophy in low gravity is infrastructure failure. The body must be maintained as precisely as any mechanical system. People come here the way they perform any other essential task.

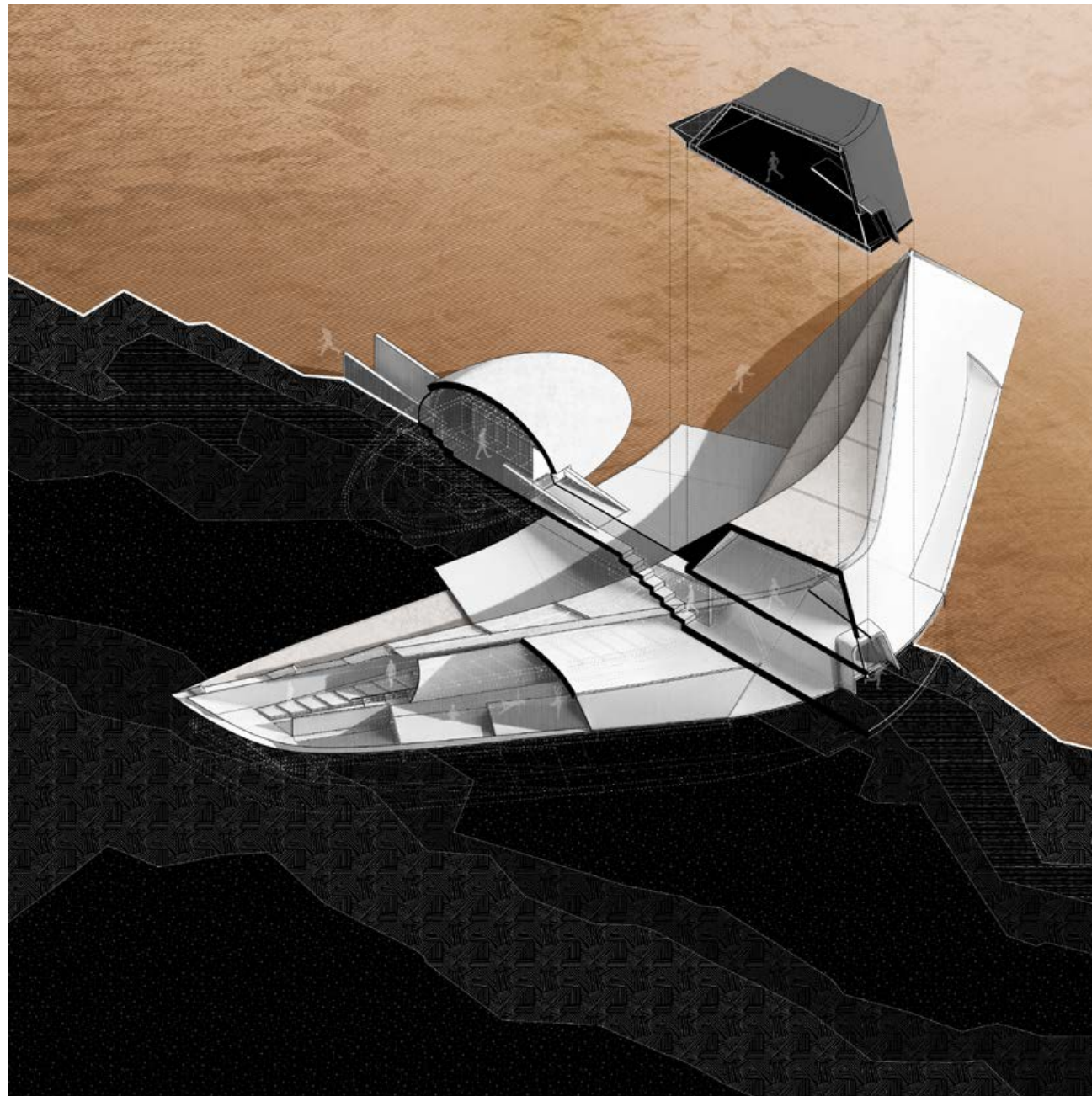
Along one wall, slot windows cut long horizontal lines just below standing eye level. The view withholds itself until you stop, until you sit or find stillness, and then the landscape opens. Flat rock. Rust-colored distance. A horizon that asks nothing and offers nothing back, and yet something in that emptiness settles the mind. People sit with it the way you might sit with water. Just locating themselves inside something larger.

The window does not frame a view. It conditions one.

The space sits at the highest point of the site, furthest from the crater floor. Stairs descend from it, so shallow they barely announce themselves, each rise low enough that the climb back feels less like effort and more like the building drawing you upward, back into its quietest space. Below, the exercise room. Above, stillness and the slot and the rust-colored nothing beyond.

At the center, a spherical volume pushes out from the wall, round and self-contained. This is where you suit up. To step outside on Mars is not a casual thing, and the architecture knows it. The sphere holds the ritual of preparation, the threshold between the breathable world inside and the one beyond, which will kill you without the right skin.

Jay Apt described how astronauts found their way to the window in free moments. Not to escape, but to locate themselves. To understand, again, where they were. That is what this room does. It holds you, and it shows you what you are held against.



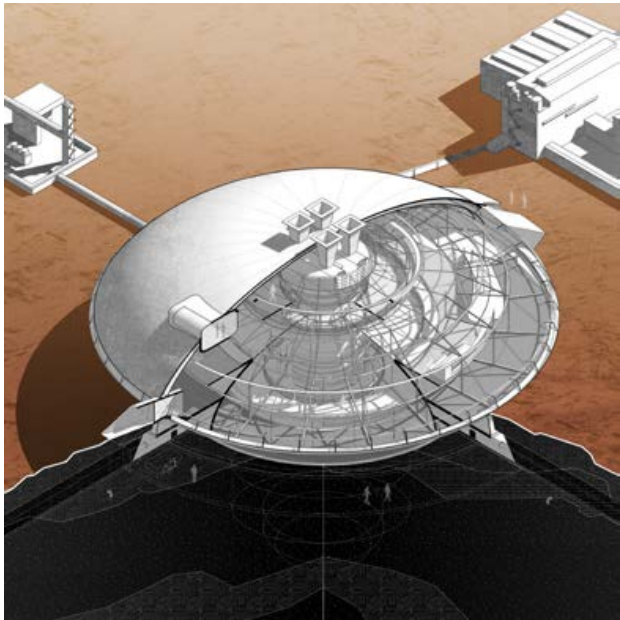
Every morning begins here. Hands in her lap, the curtain blurring the soft footsteps passing outside. Beyond it, the stillness of the landscape. This is how it has always been.

Fig. 46. Section isometric of the exercise room. Source: Author.

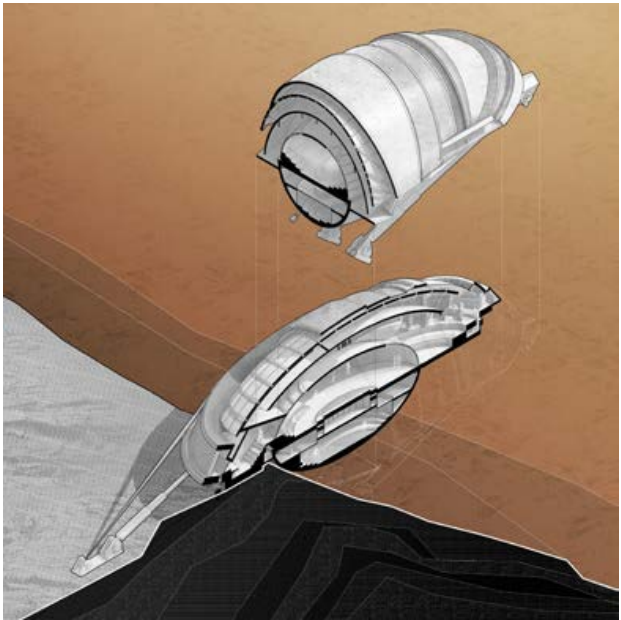


Fig. 47. Rendering of the exercise room. Source: Author.

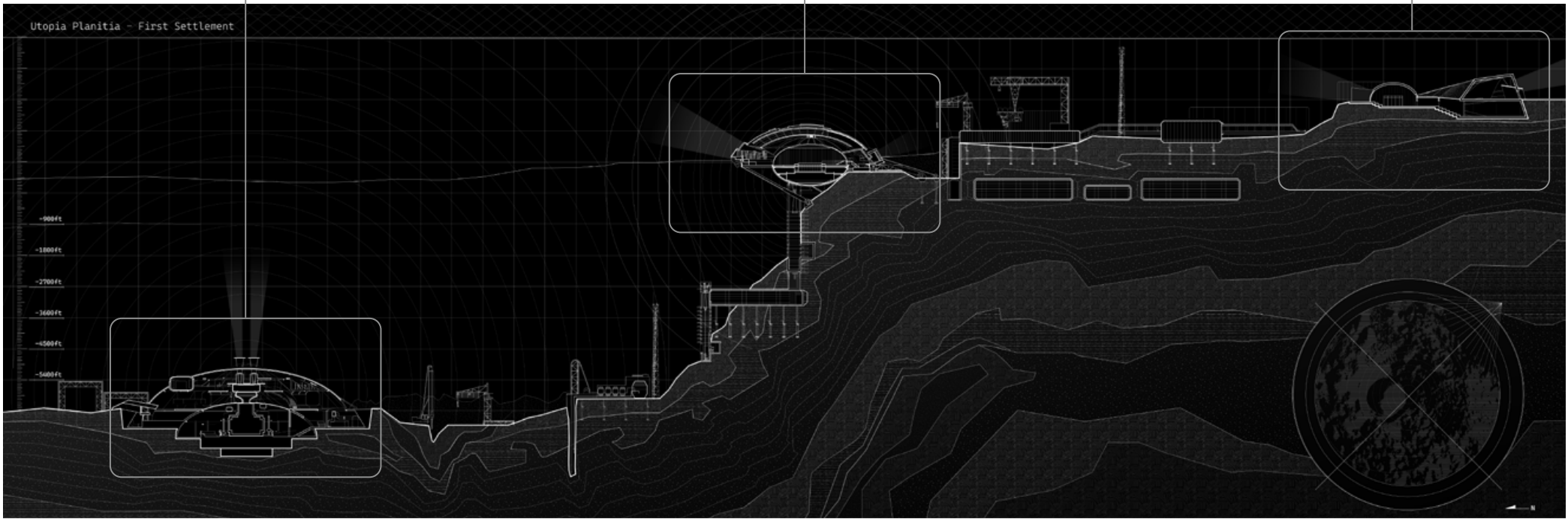
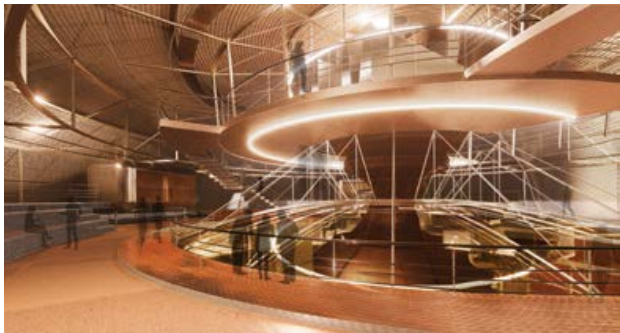
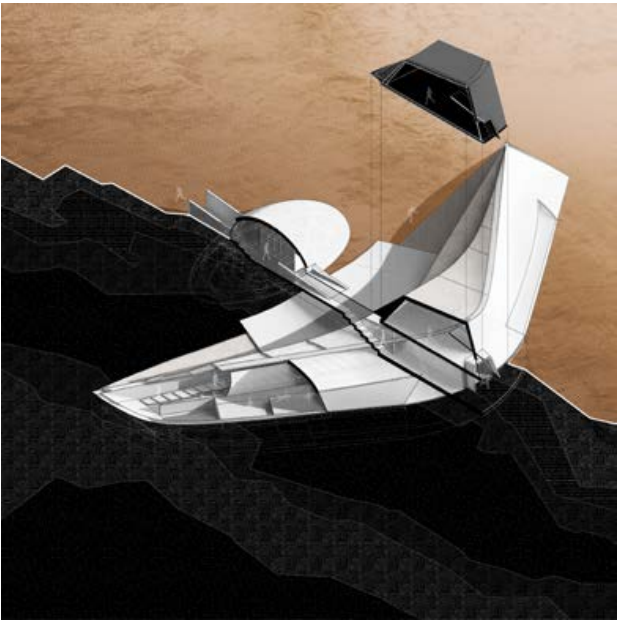
Infrastrure Hub



Greenhouse and Home



Exercise Room



Chapter 10
The Return



Fig. 48. Detailed visualization of Earth. Source: Adobe Stock, images by Brutto Film and Vadimsadovski, edited.



Mars was never the destination.

On Earth we built enough distance into our systems to make consequence invisible. Mars removes that distance entirely. When there is no frontier to expand into, no atmosphere to absorb what we take, every decision becomes local and every cost becomes visible. That is not a future condition.

Because the distance is already collapsing here on Earth.

Climate migration is already happening. People are already being pushed from places that can no longer sustain them, toward places that may not take them. The sacrifice zone is no longer somewhere else. It is moving closer. The systems we built to keep consequence at arm's length are failing, not all at once, but steadily, the way Earth frays.

We do not need to wait for Mars to ask these questions. We need to ask them now, in the cities we are already building, for the people who are already moving.

The decisions we make in architecture, about what is hidden and what is visible, what is shared and what is enclosed, are not aesthetic decisions. They are ethical ones.

Fig. 49. Earthrise, December 24, 1968. Source: William Anders, Apollo 8, NASA.



Fig. 50. Open pit mine. Source: Midkhat Izmaylov, Adobe Stock.



Fig. 51. Workers en route to an export processing zone, Mongla, Bangladesh, March 3, 2022. Source: Mahmud Hossain Opu, AP Photo.

Chapter 11

Exhibition

This thesis exhibition was presented from April 27, 2026, to May 9, 2026, in Pittsburgh, Pennsylvania. The dissertation defense took place on May 4, 2026.

This exhibition is organized around three moments within the Martian settlement that capture different qualities of life, space, and community. Each represents an essential aspect of long-term habitation: sustaining life, gathering with others, and supporting personal well-being.

The process panel documents the research, precedents, and design development that informed the project. It traces the progression from initial questions about human habitation in extreme environments to the final architectural proposal.

Three section isometrics examine these moments through the infrastructure hub, the greenhouse and communal dining area, and the meditation and exercise room. Together, they reveal how spatial organization, structure, environmental systems, and daily routines are integrated within the habitat.

The renderings visualize the experience of inhabiting these spaces, illustrating how light, material, scale, and enclosure shape life beneath the Martian surface.

The physical model focuses on the habitat's carved relationship to the surrounding rock. By highlighting moments of excavation and occupation, it reveals how the architecture is embedded within the landscape.

The VR experience allows visitors to move through the same three spaces represented in the drawings and model. Experiencing the habitat at full scale provides a deeper understanding of the conditions of life within the settlement.

Together, these elements offer multiple ways of understanding what it might mean to live, work, gather, and thrive within a permanent Martian habitat.





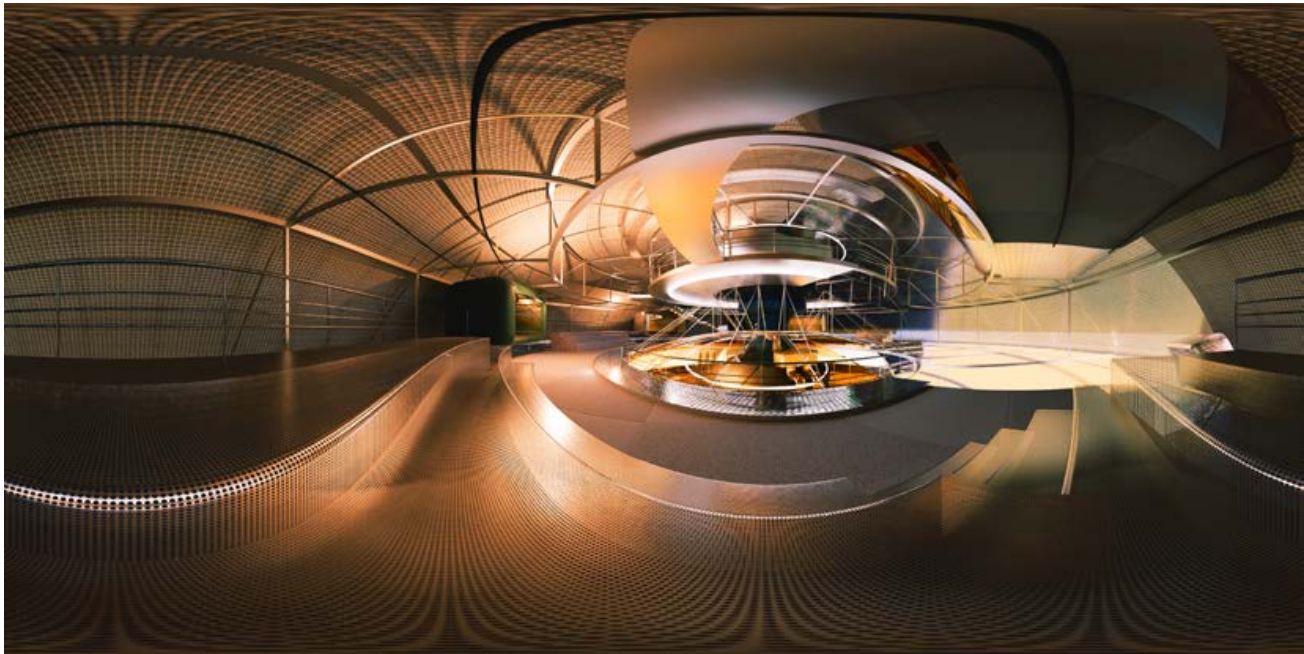


Fig. 52. VR panoramic view of the infrastructure hub. Source: Author



The 360° panoramic footage was loaded into Skybox on a Meta Quest headset, offering a stronger sense of scale and spatial layout than still images. Advisor Jared Abraham also experienced the immersive view.



Chapter 12

Reflections

This project began with an interest in speculative thinking and its ability to question the assumptions that shape our future. Inspired by the growing conversation around space exploration and Mars colonization, I initially approached the thesis as a technical design challenge. But as the project developed, I realized Mars was less a destination than a lens through which to examine larger political, cultural, and social questions.

Who gets to build a new world? If we were given the opportunity to start over, would we reproduce the same systems that have contributed to environmental degradation, inequality, and loneliness on Earth? Mars became a thought experiment for exploring these questions. In an environment where resources are finite and there is no outside to absorb consequence, architecture becomes inseparable from the systems that sustain life.

What emerged through the design process was that the most meaningful spaces were not defined by form or technology, but by the moments they made possible: gathering around a meal, looking out at the landscape, finding privacy, connection, and a sense of belonging. The project became less about designing a settlement and more about understanding what people need to live well together in conditions of extreme constraint.

The conditions explored in this thesis are often framed as distant futures, but many are already emerging around us. Climate migration, resource scarcity, and increasingly extreme environments are no longer hypothetical scenarios. They are present realities that demand new ways of thinking about how we design, build, and live together. In this sense, designing for Mars is not preparation for another planet, but a way of preparing for challenges already unfolding on Earth.

I see this thesis not as a conclusion, but as the beginning of a larger line of inquiry. In future work with Christian Ang, I hope to continue exploring how architectural frameworks and mathematical optimization can be used to imagine the growth of

human settlements over the next five hundred years. While the scale may expand, the central question remains the same: how can architecture create the conditions for people and communities to thrive in an uncertain future?

Bibliography

Aalto, Alvar. 1933. Paimio Sanatorium. Paimio, Finland.

Aalto, Alvar. 1978. Sketches. Edited by Göran Schildt. Cambridge, MA: MIT Press.

Alvar Aalto Foundation. n.d. “Paimio Sanatorium, 1928–1929.” Architecture-History.org. <https://architecture-history.org/architects/architects/aalto/objects/1928%E2%80%931929,%20Paimio%20Sanatorium,%20Tuberculosis%20sanatorium%20and%20staff%20housing,%20Paimio,%20Finland.html>.

Anderson, Douglas. 2010. “Humanizing the Hospital: Design Lessons from a Finnish Sanatorium.” CMAJ: Canadian Medical Association Journal 182 (11): E535–E537. <https://doi.org/10.1503/cmaj.090075>.

Apt, Jay. Former NASA Astronaut; Professor, Carnegie Mellon University. Personal interview, 2024.

Archigram (Peter Cook). 1964. Plug-In City.

Archigram (Ron Herron). 1964. Walking City.

architerraX. 2021. “Linear City Concept.” August 18, 2021. <https://www.architerrax.com/post/linear-city-concept>.

Archizoom. 1969. No-Stop City.

Bachelard, Gaston. 1958. The Poetics of Space. Paris: Presses Universitaires de France.

Cook, Peter, ed. 1999. Archigram. New York: Princeton Architectural Press.

Cucinotta, Francis A., Myung-Hee Y. Kim, Vicki Willingham, and Kerry A. George. 2008. “Space Radiation Cancer Risks and Uncertainties for Mars Missions.” Radiation Research 170 (1): 19–25. <https://doi.org/10.1667/RR1370.1>.

Durante, Marco, and Francis A. Cucinotta. 2011. “Physical Basis of Radiation Protection in Space Travel.” Reviews of Modern Physics 83 (4): 1245–1281. <https://doi.org/10.1103/RevModPhys.83.1245>.

Frampton, Kenneth. 2007. Modern Architecture: A Critical History. 4th ed. London: Thames & Hudson.

Gakis, D., and D. Atri. 2022. “Modeling the Effectiveness of Radiation Shielding Materials for

Astronaut Protection on Mars.” arXiv preprint. <https://arxiv.org/abs/2205.13786>.

Harvard Gazette. 2017. “Over Nearly 80 Years, Harvard Study Has Been Showing How to Live a Healthy and Happy Life.” April 11, 2017. <https://news.harvard.edu/gazette/story/2017/04/over-nearly-80-years-harvard-study-has-been-showing-how-to-live-a-healthy-and-happy-life/>.

Hofmann, N., et al. 2020. “A Review Towards the Design of Extraterrestrial Structures: From Regolith to Human Outposts.” Acta Astronautica 175: 540–569. <https://doi.org/10.1016/j.actaastro.2020.05.038>.

Ingels, Bjarke, and BIG. 2017. Mars Science City. Dubai: Mohammed bin Rashid Space Centre and Bjarke Ingels Group.

Institute of Biomedical Problems (IBMP) and European Space Agency (ESA). 2011. Mars500 Mission Overview and Results of 520-Day Isolation Experiment. https://www.esa.int/Science_Exploration/Human_and_Robotic_Exploration/Mars500.

Integrated Urbanism Studio, University of Toronto. n.d. “Quinta da Malagueira: Álvaro Siza.” John H. Daniels Faculty of Architecture, Landscape, and Design. <https://academic.daniels.utoronto.ca/urbanism/precedent-list/>.

Kim, T. G., et al. 2014–2018. Hawai’i Space Exploration Analog and Simulation (HI-SEAS) Mission Reports. Honolulu: University of Hawai’i. <https://hi-seas.org>.

Koolhaas, Rem, and Elia Zenghelis. 1972. Exodus, or the Voluntary Prisoners of Architecture.

Mies van der Rohe, Ludwig. 1937–1938. Resor House Project. Unbuilt project, Jackson Hole, Wyoming.

Mies van der Rohe, Ludwig. 1947–1951. Farnsworth House. Plano, Illinois.

Miliutin, Nikolai. 1930. Sotsgorod: The Problem of Building Socialist Cities. Translated by Arthur Sprague. Cambridge, MA: MIT Press, 2021.

Naktin, Alec, and Natalie Perri. 2020. Thresholds. First Prize Winner, Marsception Competition. Mumbai: Volume Zero Competitions.

NASA. 2019. “3D-Printed Habitat Challenge.” NASA Centennial Challenges Program. Washington, DC: National Aeronautics and Space Administration.

NASA. n.d. “Bigelow Expandable Activity Module (BEAM).” Accessed May 24, 2026. <https://www.nasa.gov/international-space-station/bigelow-expandable-activity-module-beam/>.

NASA. n.d. “Human Research Program: Behavioral Health and Performance.” <https://www.nasa.gov/hrp>.

NASA. n.d. “Utopia Planitia: Subsurface Ice and Mineral Resources.” NASA Planetary Science Division.

NASA/JPL-Caltech/University of Rome/ASI/PSI. 2016. “Location of Large Subsurface Water-Ice Deposit in Utopia Planitia, Mars.” NASA Science. November 22, 2016. <https://science.nasa.gov/resource/location-of-large-subsurface-water-ice-deposit-in-utopia-planitia-mars/>.

Nieuwenhuys, Constant. 1956–1974. New Babylon.

Pallasmaa, Juhani. 1996. *The Eyes of the Skin: Architecture and the Senses*. London: Academy Editions.

Piano, Renzo, and Richard Rogers. 1977. *Centre Georges Pompidou*. Paris.

Röstel, L., et al. 2020. “Thermal and Environmental Conditions in Subsurface Planetary Habitats.” *Planetary and Space Science* 180: 104754. <https://doi.org/10.1016/j.pss.2019.104754>.

Rudofsky, Bernard. 1987. *Architecture Without Architects: A Short Introduction to Non-Pedigreed Architecture*. Albuquerque: University of New Mexico Press.

Sadler, Simon. 2005. *Archigram: Architecture Without Architecture*. Cambridge, MA: MIT Press.

Schildt, Göran. 1994. *Alvar Aalto: The Complete Catalogue of Architecture, Design and Art*. New York: Rizzoli.

Scott, Felicity D. 2016. *Outlaw Territories: Environments of Insecurity / Architectures of Counterinsurgency*. New York: Zone Books.

Simonsen, Lisa C., John E. Nealy, Lawrence W. Townsend, and John W. Wilson. 1990. “Space Radiation Shielding for a Martian Habitat.” SAE Technical Paper. International Conference on Environmental Systems. <https://saemobilus.sae.org>.

Sinn, Thomas, and Ondrej Doule. n.d. “Inflatable Structures for Mars Base 10.” Advanced Space Concepts Laboratory / University of Strathclyde, Glasgow, and Space Innovations / International Space University, Illkirch-Graffenstaden.

Siza, Álvaro. 1977–1997. *Malagueira Housing Estate*. Évora, Portugal.

Siza, Álvaro. 1977. *Bouça Housing*. Porto, Portugal.

Soria y Mata, Arturo. 1882. *La Ciudad Lineal*. Madrid: Compañía Madrileña de Urbanización.

Talen, Mark R. 2024. “The Good Life: Lessons From the World’s Longest Scientific Study of Happiness.” *Family Medicine* 56 (10): 684–685. <https://doi.org/10.22454/FamMed.2024.345850>.

Universe Today. n.d. “Which Parts of Mars Are the Safest from Cosmic Radiation?” <https://www.universetoday.com/articles/which-parts-of-mars-are-the-safest-from-cosmic-radiation>.

[universetoday.com/articles/which-parts-of-mars-are-the-safest-from-cosmic-radiation](https://www.universetoday.com/articles/which-parts-of-mars-are-the-safest-from-cosmic-radiation).

Viúdez-Moreiras, D. n.d. “The Ultraviolet Radiation Environment and Shielding in Pit Craters and Cave Skylights on Mars.” Centro de Astrobiología (CSIC-INTA) & National Institute for Aerospace Technology (INTA), Torrejón de Ardoz, Madrid.

Wagner, M., et al. 2000. “Inflatable Structures for Space Applications.” *Acta Astronautica* 47 (2–9): 215–224. [https://doi.org/10.1016/S0094-5765\(00\)00086-1](https://doi.org/10.1016/S0094-5765(00)00086-1).

Waldinger, Robert J., and Marc S. Schulz. 2023. *The Good Life: Lessons from the World’s Longest Scientific Study of Happiness*. New York: Simon & Schuster.

Zeitlin, Cary, Donald M. Hassler, Robert F. Wimmer-Schweingruber, et al. 2013. “Measurements of Energetic Particle Radiation in Transit to Mars on the Mars Science Laboratory.” *Science* 340 (6136): 1080–1084. <https://doi.org/10.1126/science.1235989>.

Zhang, Yannis, Dong Hoon Lee, Leandro F. M. Rezende, et al. 2026. “Long-Term Resistance Training with All-Cause and Cause-Specific Mortality: Assessing Dose-Response and Joint Associations with Aerobic Physical Activity.” *British Journal of Sports Medicine* 60: 874–883.

Zumthor, Peter. 1996. *Therme Vals*. Vals, Switzerland.

Zumthor, Peter. 2006. *Atmospheres: Architectural Environments, Surrounding Objects*. Basel: Birkhäuser.

About Me



I am a fifth-year architecture student at Carnegie Mellon University interested in the relationship between architecture, technology, and human experience. Rather than seeing design as the creation of objects or forms, I see it as a way of shaping how people live, interact, and make sense of the environments around them.

My interest in housing first led me to question how architecture might respond to everyday life rather than efficiency alone. Studying abroad at EPFL expanded those questions from the scale of the home to the scale of the city, while my education at Carnegie Mellon reinforced a belief that continues to guide my work: design is never neutral. The spaces we create shape who belongs, who has access, and how communities are formed.

This thesis pushes those questions to their limit through the lens of a city on Mars. In an environment where every resource is finite and every spatial decision carries immediate collective consequences, architecture cannot be separated from the systems, values, and ways of life it produces. Mars is not the destination of this work but a lens through which to reconsider Earth. By imagining life under conditions of scarcity, interdependence, and environmental uncertainty, this project asks how architecture might help us build more resilient, equitable, and collective futures here at home.

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